

# Business Domain Study and Database Design Project
# Software Platform Domain: Netflix

NETFLIX

Netflix, Inc.


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# 1 Netflix Overview

Netflix is a global streaming service that offers a vast library of TV shows, movies, documentaries, and original content. It allows users to watch their favorite content on various devices, including TVs, computers, smartphones, and tablets. Netflix has revolutionized the way people consume entertainment by providing on-demand access to a wide range of programming.

## 1.1 Business Processes

### 1.1.1 Netflix Tudum

Netflix Tudum is an online platform created by Netflix to celebrate its content and engage with its audience. The name "Tudum" is inspired by the sound that plays when you start watching something on Netflix. It serves as a way for the streaming giant to connect with fans by providing articles containing exclusive interviews, announcements, and behind-the-scenes access to its original movies, series, and stars.

### 1.1.2 User Subscription Process

This process encompasses all the steps involved in a user signing up for a subscription, managing their account, and maintaining their subscription over time. The goal is to create a seamless and user-friendly experience that encourages potential subscribers to sign up and helps retain them for the long term.

### 1.1.3 Content Tagging Process

Netflix employs content analysts, known colloquially as “taggers” - employees whose roles are to assign specific keywords to various content on the platform. [10] These keywords include the release year, director cast, theme, language, how much sex or profanity is in the

content and genre of said content. These keywords are then used as metadata to help promote content to the right audience.

### 1.1.4 Netflix Shop

**Netflix Shop** is an online store where you can find merchandise inspired by your favorite Netflix shows and movies. It offers a variety of products, including:

**Apparel:** T-shirts, hoodies, hats, and other clothing items featuring characters, logos, and quotes from popular Netflix series.

**Collectibles:** Figures, toys, and other collectible items based on your favorite shows.

**Home Goods:** Decorative items, such as mugs, blankets, and wall art, inspired by Netflix content.

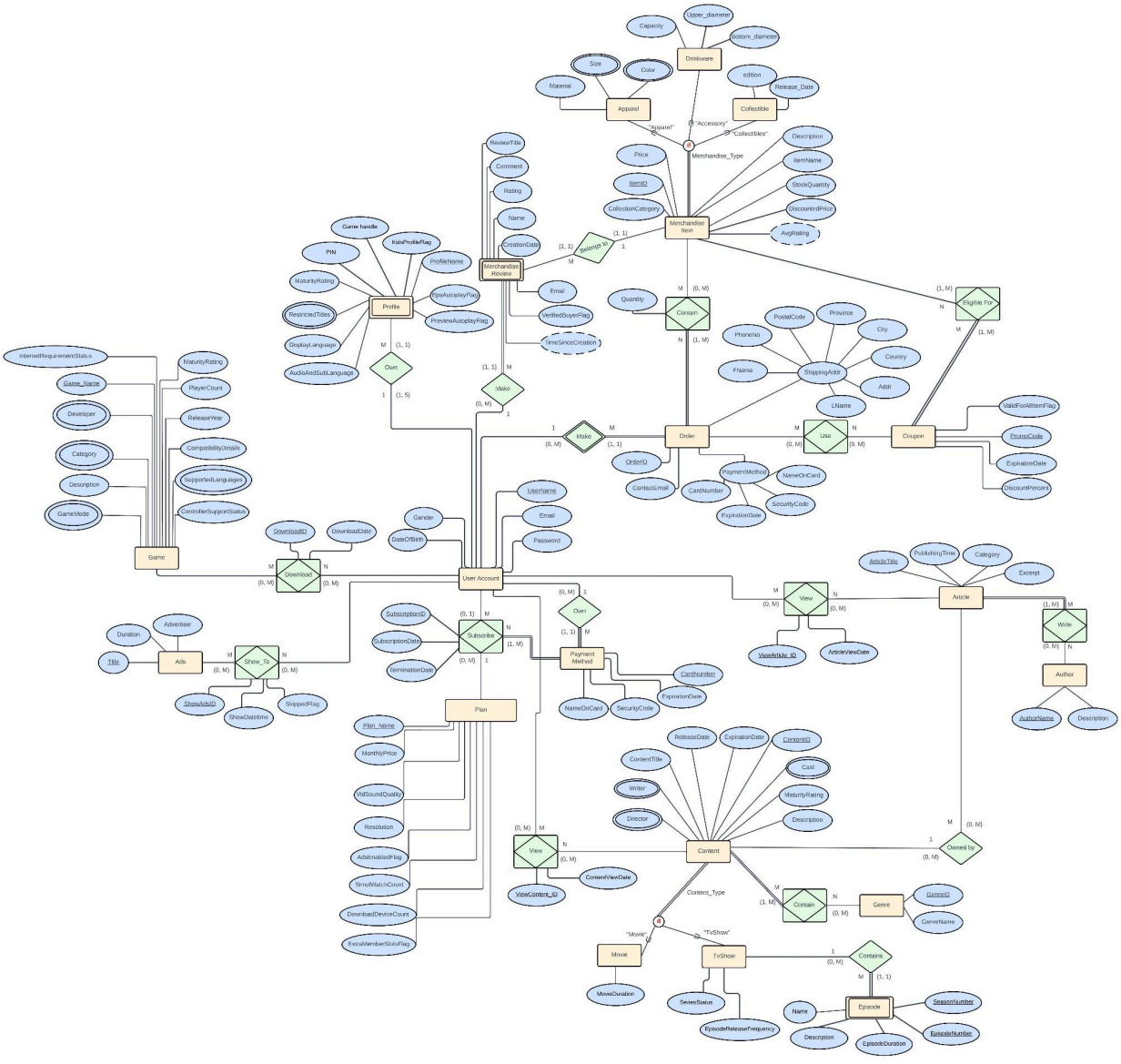
### 1.1.5 Netflix Games

Netflix Games is a gaming platform integrated within the Netflix app that allows subscribers to play a variety of mobile games on their smartphones and tablets. The games cover various genres, including action, puzzle, adventure, and strategy, and are designed to cater to different age groups and preferences. The games are free from ads, in-app purchases, and additional costs, making them accessible to all Netflix subscribers as part of their existing membership. Users must download the games before playing, making Netflix Games similar in functionality to game publishing platforms, such as Google Play.

### 1.1.6 Advertising Process

Netflix Advertising, introduced in 2022 with the "Basic with Ads" plan, is the company's venture into offering ad-supported streaming. This service allows users to subscribe at a lower cost while viewing ads during shows and movies.

## 1.2 ERD


Capacity
Upper_diameter
Bottom_diameter
Drinkware
Size
Color
Material
Apparel
edition
Release_Date
Collectible
"Apparel"
"Accessory"
"Collectibles"
Merchandise_Type
Price
ItemID
CollectionCategory
Description
ItemName
StockQuantity
DiscountedPrice
AvgRating
Merchandise Item
ReviewTitle
Comment
Rating
Name
CreationDate
Merchandise Review
Belongs to
(1, 1)
(1, M)
Game handle
KidsProfileFlag
PIN
ProfileName
MaturityRating
RestrictedTitles
DisplayLanguage
AudioAndSubLanguage
TipsAutoplayFlag
PreviewsAutoplayFlag
Profile
Email
VerifiedBuyerFlag
TimeSinceCreation
Own
Make
Quantity
Contain
Eligible For
PostalCode
Province
City
Country
PhoneNo
FName
ShippingAddr
Addr
LName
Order
OrderID
ContactsEmail
CardNumber
PaymentMethod
NameOnCard
SecurityCode
ExpirationDate
Use
Coupon
ValidForAllItemFlag
PromoCode
ExpirationDate
DiscountPercent
InternetRequirementsStatus
Game_Name
Developer
Category
Description
GameMode
MaturityRating
PlayerCount
ReleaseYear
CompatibilityDetails
SupportedLanguages
ControllerSupportStatus
Game
DownloadID
DownloadDate
Download
Gender
DateOfBirth
UserName
Email
Password
User Account
View
ViewArticle_ID
ArticleViewDate
Article
ArticleTitle
PublishingTime
Category
Excerpt
Write
Author
AuthorName
Description
Duration
Advertiser
Title
Ads
Show_To
ShowVidID
ShowDatetime
SkippedFlag
SubscriptionID
SubscriptionDate
TerminationDate
Subscribe
Own
Payment Method
CardNumber
ExpirationDate
NameOnCard
SecurityCode
Plan
Plan_Name
MonthlyPrice
VidSoundQuality
Resolution
AdsEnabledFlag
SimultWatchCount
DownloadDeviceCount
ExtraMemberSlotsFlag
ReleaseDate
ExpirationDate
ContentID
Cast
MaturityRating
Description
ContentTitle
Writer
Director
Content
Owned by
View
ViewContent_ID
ContentViewDate
Content_Type
"Movie"
"TvShow"
Movie
MovieDuration
TvShow
SeriesStatus
EpisodeReleaseFrequency
Contains
Episode
Name
Description
EpisodeDuration
SeasonNumber
EpisodeNumber
Contain
Genre
GenreID
GenreName

## 2.Step-by-Step Relational Diagram Transformation

### Step1: Strong Entity

**Game**

| Game_Name | Description | InternetRequirementStatus | MaturityRating | PlayerCount | ReleaseYear | ControllerSupportStatus | CompatibilityDetails |
|---|---|---|---|---|---|---|---|

**User Account**

| UserName | Email | Password | DateOfBirth | Gender |
|---|---|---|---|---|

**Plan**

| Plan_Name | MonthlyPrice | VidSoundQuality | Resolution | AdsEnabledFlag | SimulWatchCount | DownloadDeviceCount | ExtraMemberSlotsFlag |
|---|---|---|---|---|---|---|---|

**Payment Method**

| CardNumber | ExpirationDate | NameOnCard | SecurityCode |
|---|---|---|---|

**Content**

| ContentTitle | ReleaseDate | ExpirationDate | ContentID | MaturityRating | Description | TotalDuration |
|---|---|---|---|---|---|---|

**Genre**

| GenreID | GenreName |
|---|---|

**Article**

| ArticleTitle | PublishingTime | Category | Excerpt |
|---|---|---|---|

**Author**

| AuthorName | Description |
|---|---|

**Order**

| OrderID | ContactEmail | CardNumber | ExpirationDate | SecurityCode | NameOnCard | FName | LName | PostalCode | Province | City | Country | Addr |
|---|---|---|---|---|---|---|---|---|---|---|---|---|

**Merchandise Item**

| CollectionCategory | ItemID | Price | Description | ItemName | StockQuantity | DiscountedPrice |
|---|---|---|---|---|---|---|

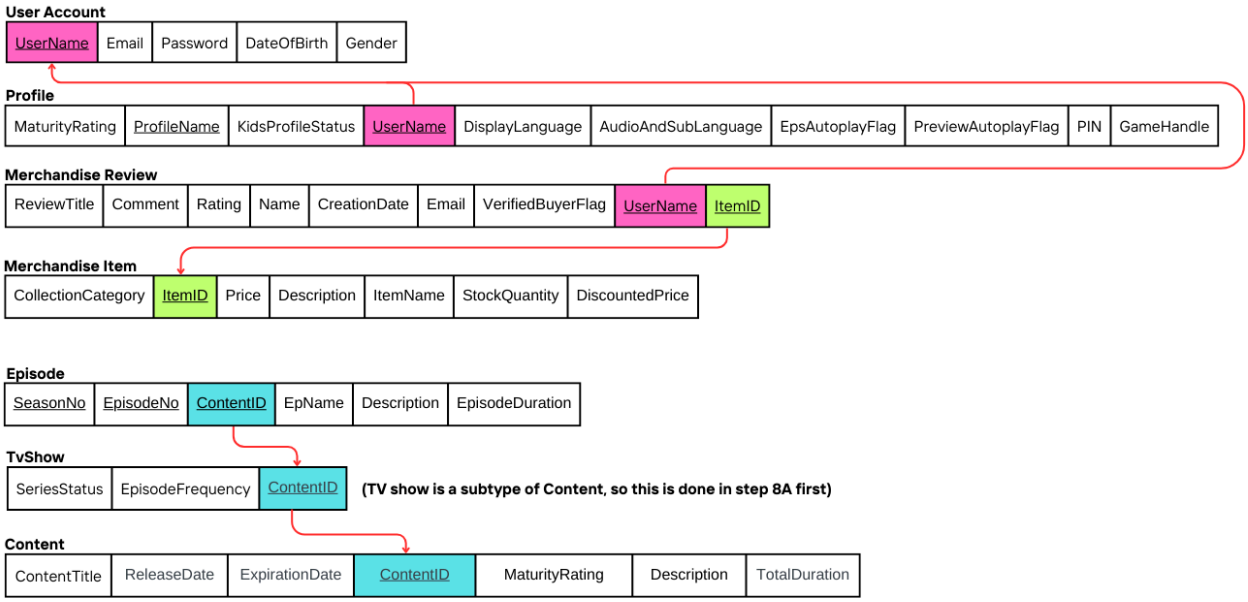
**Ads**

| AdsTitle | Duration | Advertiser |
|---|---|---|

**Coupon**

| PromoCode | ExpirationDate | DiscountPercent | IsValidForAllItem |
|---|---|---|---|

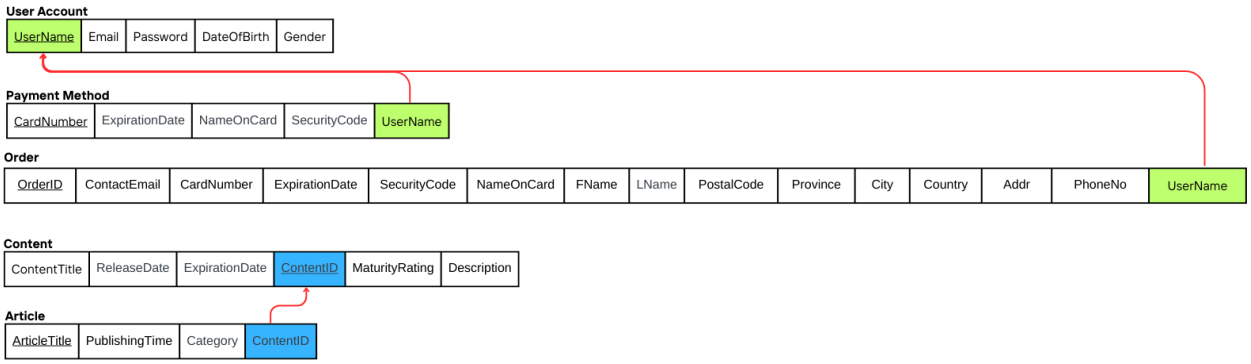
## Step2: Weak Entity



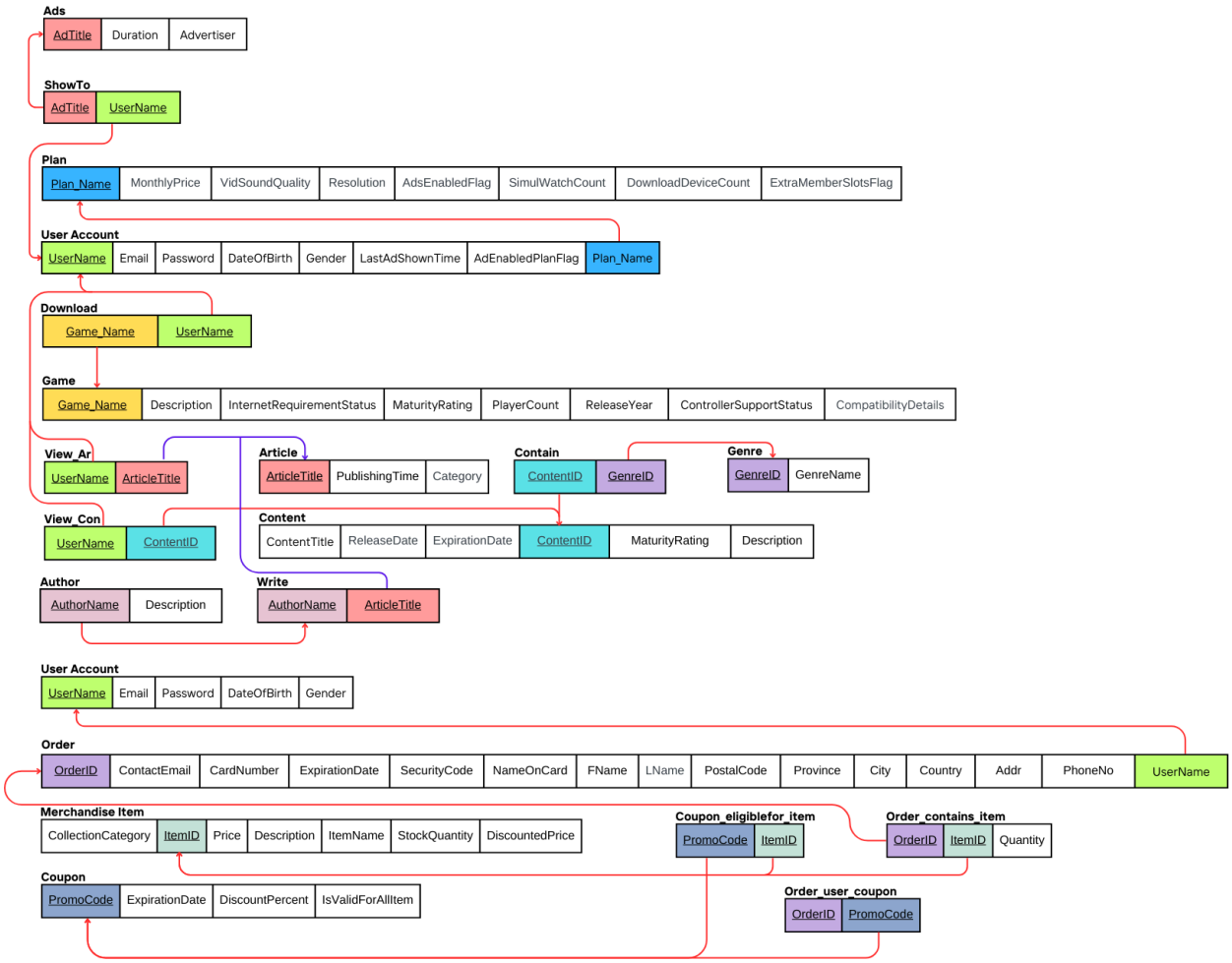

## Step3: Binary 1:1 Relationship

There exists no 1:1 relationship in this schema

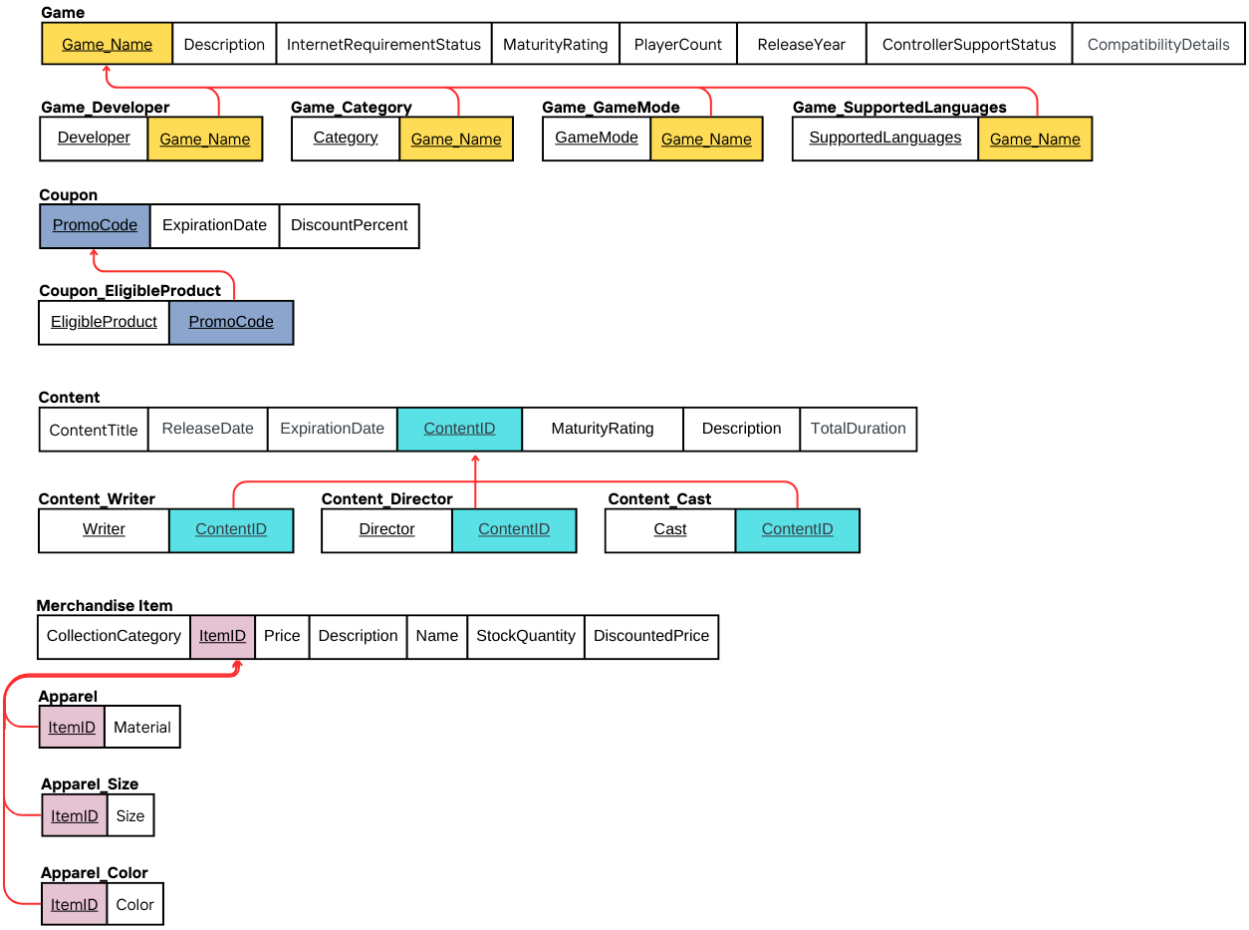
## Step4: Binary 1:M Relationship



## Step5: Binary M:N Relationship



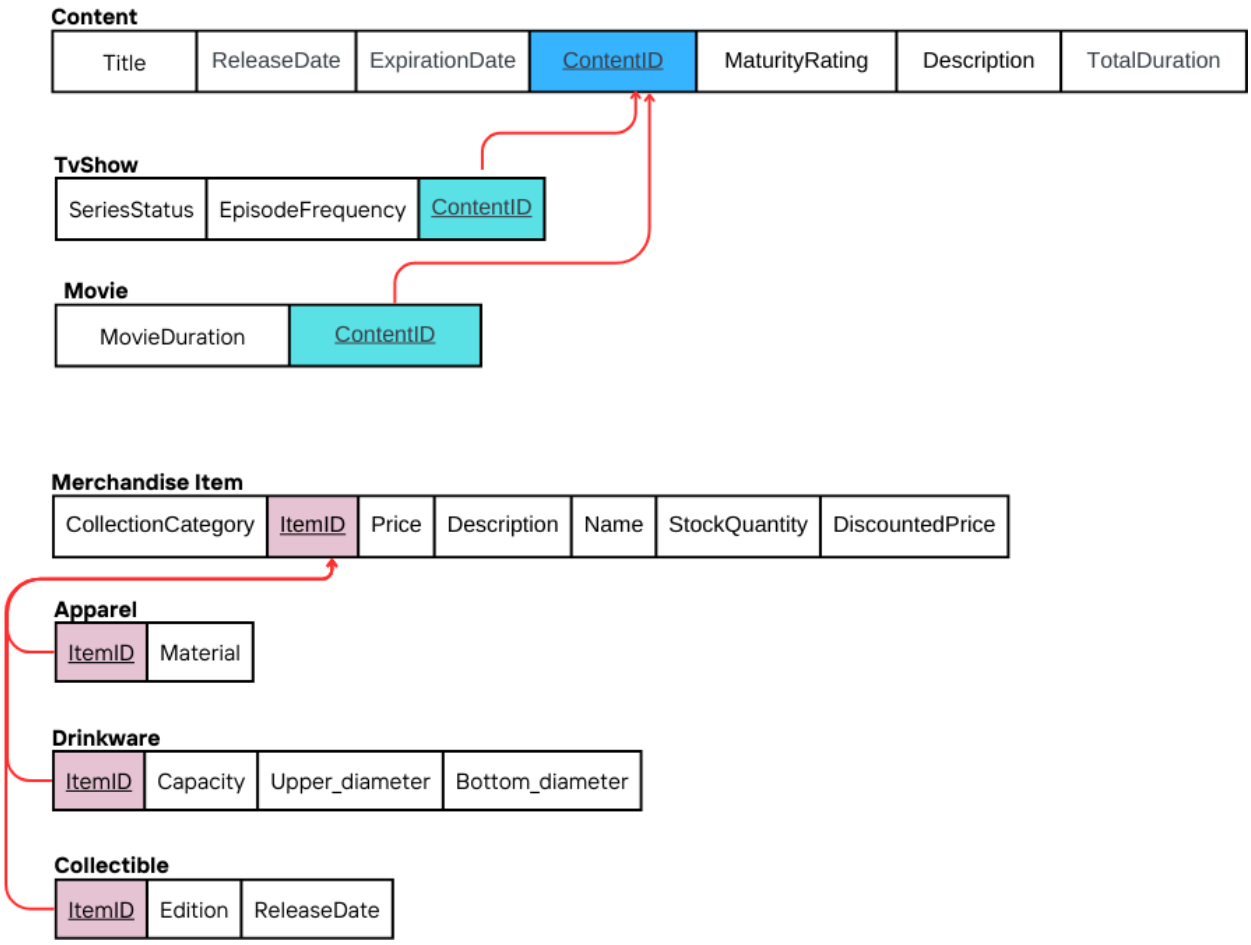
## Step6:Multi-Valued Attribute



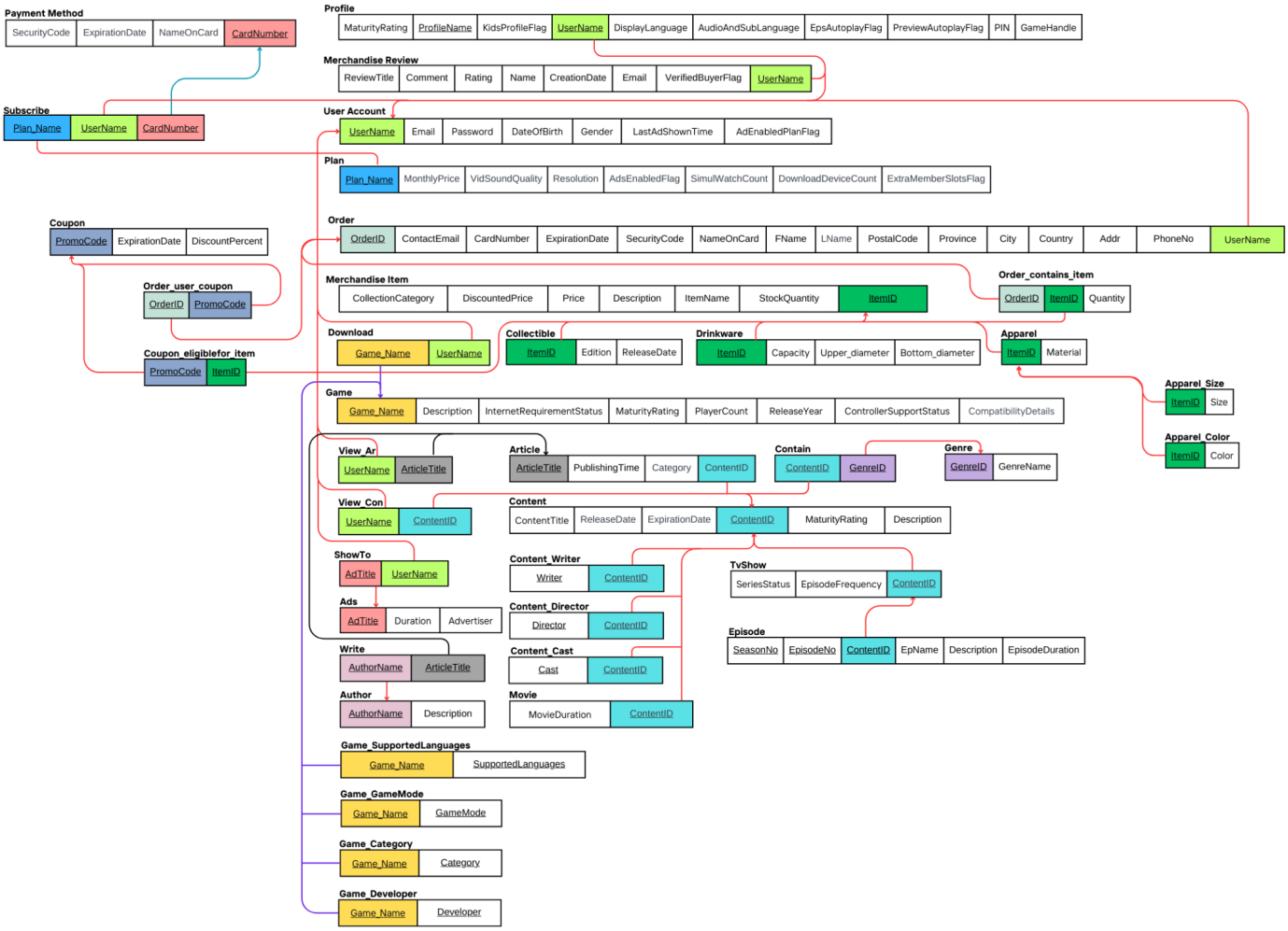

## Step7: N-ary Relationship

There exists no N-ary relationship in this schema

## Step8A: Specialization



# Final Relational Schema



# 3 Data Dictionary

| Table Name | Attribute Name | Contents | Type | Format | Nullable | Range | Key | FK Referenced Table |
|---|---|---|---|---|---|---|---|---|
| ads | ads_title | The title of an ad. | varchar(20) | xxxxxxx | | | PK | |
| | duration | The ad's duration, measured in seconds. | int(11) | x | | 0 to 60 | | |
| | advertiser | The advertiser of this ad. | varchar(20) | xxxxxxx | | | | |
| apparel | item_id | Item ID. | int(11) | x | | | PK,FK | item_id [merchandise_item] |
| | material | Description of the material the item is made of. | varchar(50) | xxxxxxx | | | | |
| apparel_color | item_id | Item ID | int(11) | x | | | PK,FK | item_id [merchandise_item] |
| | color | Available color of the item. | varchar(10) | xxxxxxx | | | PK | |
| apparel_size | item_id | Item ID | int(11) | x | | | PK,FK | item_id [apparel] |
| | size | Available size of the apparel. | varchar(4) | xxxx | | | PK | |
| article_tudum | article_title | An article's title | varchar(50) | xxxxxxx | | | PK | |
| | publishing_time | The datetime the article is published on. | datetime | YYYY-MM-DD hh:mm:ss | | | | |
| | category | The category of an article. | varchar(10) | xxxxxxx | | | | |
| | excerpt | Brief excerpt of the article. | varchar(300) | xxxxxxx | | | | |
| | content_id | Content ID | int(11) | x | | | FK | content_id [content] |
| author | author_name | An author's name. | varchar(50) | xxxxxxx | | | PK | |
| | description | The description of an author. | varchar(300) | xxxxxxx | | | | |
| author_write_article | author_name | Author names | varchar(50) | xxxxxxx | | | PK,FK | author_name [author] |
| | article_title | Article title | varchar(100) | xxxxxxx | | | PK,FK | article_title [article_tudum] |
| collectible | item_id | Item ID. | int(11) | x | | | PK,FK | item_id [merchandise_item] |
| | edition | Edition number of the collection. | int(2) | x | | | | |
| | release_date | Release date of the collection. | date | YYYY-MM-DD | | | | |
| content | content_id | Content ID | int(11) | x | | | PK | |
| | content_title | Content title | varchar(70) | xxxxxxx | | | | |
| | release_date | Release date | date | YYYY-MM-DD | | | | |
| | expiration_date | date | date | YYYY-MM-DD | Y | | | |
| | maturity_rating | Maturity rating | varchar(5) | xxxxxxx | | | | |
| | description | Description | text | xxxxxxx | | | | |
| content_cast | content_id | Content ID | int(11) | x | | | PK,FK | content_id [content] |
| | cast | Cast name | varchar(50) | xxxxxxx | | | PK | |
| content_director | content_id | Content ID | int(11) | x | | | PK,FK | content_id [content] |
| | director | Director name | varchar(50) | xxxxxxx | | | PK | |
| content_writer | content_id | Content ID | int(11) | x | | | PK,FK | content_id [content] |
| | writer | Writer name | varchar(50) | xxxxxxx | | | PK | |
| coupon | promo_code | Promotion code. | varchar(10) | xxxxxxx | | | PK | |
| | expiration_date | Coupon expiration date. | date | YYYY-MM-DD | | | | |
| | discount_percent | Discount percent offered by coupon. | int(3) | x | | | | |
| | is_valid_for_all_items | Mark if coupon is valid for all items. | tinyint(1) | x | | | | order_id [merchandise_order] |
| coupon_eligiblefor_item | promo_code | Coupon's promo code | varchar(10) | xxxxxxx | | | PK,FK | promo_code[coupon] |
| | item_id | Merchandise item ID. | int(20) | x | | | PK,FK | item_id [merchandise_item] |
| drinkware | item_id | Item ID | int(11) | x | | | PK,FK | item_id [merchandise_item] |
| | capacity | Capacity of drinkware, measured in ml. | int(5) | x | | | | |
| | upper_diameter | Upper diameter of drinkware, measured in inches. | int(2) | x | | | | |
| | bottom_diameter | Bottom diameter of drinkware, measured in inches. | int(2) | x | | | | |
| game | game_name | A game's name | varchar(20) | xxxxxxx | | | PK | |
| | description | Game's description | text | x | | | | |
| | internet_requirement_status | Indicates if internet is required to play. | tinyint(1) | x | | | | |
| | maturity_rating | Maturity rating | varchar(4) | xxxx | | | | |
| | player_count | Min/max amount of players required to play. | varchar(5) | xxxxx | | | | |
| | release_year | Release year | year(4) | YYYY | | | | |
| | controller_support_status | Indicates if controller is supported. | tinyint(1) | x | | | | |
| | compatibility_details | Device specs needed to play. | text | xxxxxxx | | | | |

| | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| game_category | game_name | Game name | varchar(50) | xxxxxxx | | | PK,FK | game_name [game] |
| | category | Game Category | varchar(20) | xxxxxxx | | | PK | |
| game_developer | game_name | Game Name | varchar(50) | xxxxxxx | | | PK,FK | game_name [game] |
| | developer | The game's developer | varchar(20) | xxxxxxx | | | PK | |
| game_gamemode | game_name | Game name | varchar(50) | xxxxxxx | | | PK,FK | game_name [game] |
| | game_mode | Different game modes of the game. | varchar(20) | xxxxxxx | | | PK | |
| game_supported_languages | game_name | Game name | varchar(50) | xxxxxxx | | | PK,FK | game_name [game] |
| | supported_lang | Supported languages of the game. | varchar(20) | xxxxxxx | | | PK | |
| genre | genre_id | Genre ID | int(11) | x | | | PK | |
| | genre_name | Genre Name | varchar(30) | xxxxxxx | | | | |
| genre_contain_content | genre_id | Genre ID | int(11) | x | | | PK,FK | genre_id [genre] |
| | content_id | Content ID | int(11) | x | | | PK,FK | content_id [content] |
| merchandise_item | item_id | Item ID | int(11) | x | | | PK | |
| | collection_category | Collection category the item belongs to. | varchar(50) | xxxxxxx | | | | |
| | price | Price | decimal(10,2) | xxxxxxx | | | | |
| | description | Description | varchar(300) | x | | | | |
| | item_name | Item Name | varchar(50) | xxxxxxx | Y | | | |
| | discounted_price | Discounted price. | decimal(10,2) | xxxxxxx.xx | | | | |
| | stock_quantity | Stock quantity | int(11) | x | | | | |
| merchandise_order | order_id | Order's ID | int(11) | x | | | PK | |
| | username | User name | varchar(20) | xxxxxxx | | | FK | username [user_account] |
| | contact_email | Contact Email | varchar(254) | xxxxxxx | | | | |
| | card_number | Credit card number | varchar(19) | xxxxxxx | | | | |
| | expiration_date | Expiration date on credit card | char(5) | xxxxx | | | | |
| | security_code | Security code on credit card | char(3) | xxx | | | | |
| | name_on_card | Name on credit card | varchar(26) | xxxxxxx | | | | |
| | fname | First Name | int(50) | x | | | | |
| | lname | Last Name | varchar(50) | xxxxxxx | | | | |
| | postal_code | Postal Code | varchar(10) | xxxxxxx | | | | |
| | province | Province | varchar(20) | xxxxxxx | | | | |
| | city | City | varchar(20) | xxxxxxx | | | | |
| | country | Country | varchar(20) | xxxxxxx | | | | |
| | addr | Extra address information | varchar(200) | xxxxxxx | Y | | | |
| | order_date | Creation date of the order | date | YYYY-MM-DD | | | | |
| | status | Order completion status | varchar(10) | xxxxxxx | | Shipping, Cancelled, Completed | | |
| merchandise_review | review_id | Review ID | int(11) | xxxxxxx | | | PK | |
| | username | User name | varchar(20) | xxxxxxx | | | PK,FK | username [user_account] |
| | item_id | Item ID | int(11) | x | | | PK,FK | item_id [merchandise_item] |
| | review_title | Review Title | varchar(50) | xxxxxxx | | | | |
| | comment | Textual content of the review | varchar(200) | xxxxxxx | | | | |
| | rating | Rating given | int(1) | x | | | | |
| | name | Displayed name on the review | varchar(20) | xxxxxxx | | | | |
| | creation_date | Creation date | date | YYYY-MM-DD | | | | |
| | email | Email | varchar(254) | xxxxxxx | | | | |
| | verified_buyer_flag | Indicates if user is a verified buyer. | tinyint(1) | x | | | | |
| movie | content_id | Content ID | int(11) | x | | | PK,FK | content_id [content] |
| | movieduration | Movie duration | time | HH:MM:SS | | | | |
| order_contains_item | order_id | Order's ID | int(11) | x | | | PK,FK | order_id [merchandise_order |
| | item_id | The ID of the merchandise item | int(11) | x | | | PK,FK | item_id [merchandise_item] |
| | quantity | The quantity of the item in the order | int(11) | x | | | | |
| order_use_coupon | order_id | Order's ID | int(11) | x | | | PK,FK | order_id [merchandise_order |
| | promo_code | The coupon code used in the order | varchar(10) | xxxxxxx | | | PK,FK | promo_code [coupon] |

| | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| payment_method | card_number | Credit Card Number | varchar(19) | xxxxxxx | | | PK | |
| | username | Username of the user owning this payment payment method | varchar(20) | xxxxxxx | | | FK | username [user_account] |
| | expiration_date | Expiration date of credit card | char(5) | xxxxx | | | | |
| | security_code | Security code of credit card. | char(3) | xxx | | | | |
| | name_on_card | Name on credit card. | varchar(26) | xxxxxxx | | | | |
| plan | plan_name | Plan name | char(10) | xxxxx | | | PK | |
| | monthly_price | Monthly price | decimal(4,2) | xxxx.xx | | | | |
| | vid_sound_quality | Video and sound quality level | varchar(15) | xxxx | | | | |
| | resolution | Content resolution | varchar(10) | xxxxxxx | | | | |
| | ads_enabled_flag | Mark if plan is ads enabled | tinyint(1) | x | | | | |
| | simul_watch_count | Count how many devices are allowed to watch on this account | int(1) | x | | | | |
| | download_device_count | Count how many devices are allowed to download content using this account | int(1) | x | | | | |
| | extra_member_slots | Count how many other user accounts you can invite to become extra members | int(1) | x | | | | |
| profile | profile_name | Profile name | varchar(10) | xxxxxxx | | | PK | |
| | username | Username | varchar(20) | xxxxxxx | | | PK,FK | username [user_account] |
| | maturity_rating | Maturity rating for profile content | varchar(3) | xxxxxxx | | | | |
| | kids_profile_status | Flag for kids-specific profile | tinyint(1) | x | | | | |
| | display_language | Display language for user interface | varchar(3) | xxx | | | | |
| | audio_sub_language | Preferred audio and subtitle language | varchar(10) | xxxxxxx | | | | |
| | eps_autoplay_flag | Flag for episode autoplay in series | tinyint(1) | x | | | | |
| | preview_autoplay_flag | Flag for autoplay of content previews | tinyint(1) | x | Y | | | |
| | pin | PIN for profile access | int(6) | x | Y | | | |
| | game_handle | Game handle for Netflix Games | varchar(20) | xxxxxxx | | | | |
| subscription | subscription_id | Subscription ID | int(11) | x | | | PK | |
| | plan_name | Plan name | char(10) | xxxxxxx | | | FK | plan_name [plan] |
| | username | User name | varchar(20) | xxxxxxx | | | FK | username [user_account] |
| | card_number | Card number | varchar(19) | xxxxxxx | | | FK | card_number [payment_method] |
| | subscription_date | Subscription date | date | YYYY-MM-DD | | | | |
| | termination_date | Termination date. (Null if subscription is still active) | date | YYYY-MM-DD | Y | | | |
| tv_episode | content_id | Content ID of the episode | int(11) | x | | | PK,FK | content_id [tv_show] |
| | season_no | The number of season the episode belongs to | int(11) | x | | | PK | |
| | ep_no | The number of the episode | int(11) | x | | | PK | |
| | ep_name | An episode's name | varchar(100) | xxxxxxx | | | | |
| | description | The episode"s description | varchar(300) | xxxxxxx | | | | |
| | episode_duration | An episode's duration in seconds | int(11) | x | | | | |
| tv_show | content_id | Content ID representing the TV show | int(11) | x | | | PK,FK | content_id [content] |
| | series_status | The current status of the series | varchar(20) | xxxxxxx | | | | |
| | episode_release_frequency | Frequency of episode releases | varchar(20) | xxxxxxx | | | | |
| user_account | username | Username of the account holder | varchar(20) | xxxxxxx | | | PK | |
| | email | Email associated with the account | varchar(254) | xxxxxxx | | | | |
| | password | Hashed password | int(64) | x | | | | |
| | date_of_birth | Date of birth of the account holder | date | YYYY-MM-DD | | | | |
| | gender | Gender of the user | varchar(1) | X | Y | M,F | | |
| user_download_game | download_id | Download ID | int(11) | x | | | PK | |
| | game_name | Game name | varchar(50) | xxxxxxx | | | FK | game_name [game] |
| | username | User name | varchar(20) | xxxxxxx | | | FK | username [user_account] |
| | download_date | The date the game is downloaded by the user | date | YYYY-MM-DD | | | | |
| user_shown_ads | showads_id | A unique identifier for each ad shown. | int(11) | xxxxxxx | | | PK | |
| | ads_title | The title of the advertisement shown to the user. | varchar(20) | xxxxxxx | | | FK | ads_title [ads] |
| | user_name | The name of the user who was shown the ad. | varchar(20) | xxxxxxx | | | FK | username [user_account] |
| | show_datetime | The date and time when the ad was displayed to the user. | datetime | YYYY-MM-DD | | | | |
| | is_skipped | Indicates if the ad was skipped by the user | tinyint(1) | x | | 1,0 | | |

| | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| user_view_article | viewarticle_id | View article ID | int(11) | x | | | PK | |
| | username | User name | varchar(20) | xxxxxxx | | | FK | username [user_account] |
| | article_title | Article title | varchar(100) | xxxxxxx | | | FK | article_title [article_tudum] |
| | article_view_date | The date the article is viewed | date | YYYY-MM-DD | | | | |
| user_view_content | viewcontent_id | View Content ID | int(11) | x | | | PK,FK | username [user_account] |
| | username | User name | varchar(20) | xxxxxxx | | | FK | |
| | content_id | Content ID | int(11) | x | | | FK | content_id [content] |
| | content_view_date | The date the content is viewed | date | YYYY-MM-DD | | | | |

# 4 Table Description & Implementation

## 4.1 Ads Table

The ads table holds information about the advertisement clips that are shown to Netflix users with an ad-enabled plan subscription.

### 4.1.1 Ads Table DDL script

```
CREATE TABLE ads (
  ads_title varchar(20) NOT NULL COMMENT 'The title of an ad.',
  duration int(11) NOT NULL COMMENT 'The ad''s duration, measured in seconds.',
  advertiser varchar(20) NOT NULL COMMENT 'The advertiser of this ad.'
) ENGINE=InnoDB DEFAULT CHARSET=utf8;
ALTER TABLE ads
  ADD PRIMARY KEY (ads_title);
```

### 4.1.2 Ads table example data

```
INSERT INTO ads (ads_title, duration, advertiser) VALUES
('Electric Cars', 30, 'EcoMotors'),('Gadget Promotion', 5, 'TechWorld'),('Healthy Snacks', 10, 'Foodies'),
('Holiday Travel', 60, 'GoGlobalTrips'),('New Movie Release', 30, 'FilmStudio'),
('Streaming Deals', 60, 'StreamPlus'),('Summer Collection', 45, 'FashionHub'),
('Super Sale', 10, 'ShopNow'),('Tech Innovations', 45, 'InnovaTech'),
('Workout Gear', 5, 'FitStore');
```

## 4.2 Apparel Table

The apparel table is a specialization of merchandise items table. It holds information about the clothing type of items offered on the Netflix Shop.

### 4.2.1 Apparel table DDL script

```
CREATE TABLE apparel (
  item_id int(11) NOT NULL COMMENT 'Item ID.',
```

```
  material varchar(50) NOT NULL COMMENT 'Description of the material
the item is made of.'
) ENGINE=InnoDB DEFAULT CHARSET=utf8;
ALTER TABLE apparel
  ADD PRIMARY KEY (item_id);
  ADD KEY fk_apparel_itemid (item_id);
  ADD CONSTRAINT fk_apparel_itemid FOREIGN KEY (item_id)
REFERENCES merchandise_item (item_id);
```

#### 4.2.2 Apparel table example data

```
INSERT INTO apparel (item_id, material) VALUES
(1, '100% Cotton'), (2, 'Polyester/Cotton Blend'), (3, 'Organic Cotton'), (4,
'Polyester');
```

### 4.3 Apparel_color Table

The apparel_color table is a table representing the apparel items' colors as a multivalued attribute.

#### 4.3.1 Apparel_color Table DDL script

```
CREATE TABLE apparel_color (
 item_id int(11) NOT NULL COMMENT 'Item ID',
 color varchar(10) NOT NULL COMMENT 'Available color of the item.'
) ENGINE=InnoDB DEFAULT CHARSET=utf8;
ALTER TABLE apparel_color
  ADD PRIMARY KEY (item_id,color);
  ADD CONSTRAINT fk_color_itemid FOREIGN KEY (item_id)
REFERENCES merchandise_item (item_id);
```

#### 4.3.2 Apparel_color table example data

```
INSERT INTO apparel_color (item_id, color) VALUES
(1, 'Black'), (1, 'Red'), (2, 'Black'), (2, 'Gray'), (3, 'Blue'), (3, 'White'), (4,
'Green'), (4, 'Yellow');
```

## 4.4 Apparel_size Table

The apparel_size table is a table representing the apparel items' available sizes as a multivalued attribute.

### 4.4.1 Apparel_size Table DDL script

```
CREATE TABLE apparel_size (
 item_id int(11) NOT NULL COMMENT 'Item ID',
size varchar(4) NOT NULL COMMENT 'Available size of the apparel.'
) ENGINE=InnoDB DEFAULT CHARSET=utf8;
ALTER TABLE apparel_size
  ADD PRIMARY KEY (item_id,size);
  ADD CONSTRAINT fk_size_itemid FOREIGN KEY (item_id)
REFERENCES apparel (item_id);
```

### 4.4.2 Apparel_size table example data

```
INSERT INTO apparel_size (item_id, size) VALUES
(1, 'L'),(1, 'M'),(1, 'S'),(2, 'L'),(2, 'M'),(2, 'XL'),(3, 'L'),(3, 'M'),
(3, 'S'),(4, 'L'),(4, 'XL');
```

## 4.5 Article_tudum Table

The article_tudum table holds information about the written articles featured on Netflix Tudum, Netflix's online magazine service.

### 4.5.1 Article_tudum Table DDL script

```
CREATE TABLE article_tudum (
article_title varchar(100) NOT NULL COMMENT 'An article''s title',
publishing_time datetime NOT NULL COMMENT 'The datetime the article is published on.',
 category varchar(20) NOT NULL COMMENT 'The category of an article.',
 excerpt varchar(300) NOT NULL COMMENT 'Brief excerpt of the article.',
 content_id int(11) NOT NULL COMMENT 'Content ID'
) ENGINE=InnoDB DEFAULT CHARSET=utf8;
ALTER TABLE apparel_size
  ADD PRIMARY KEY (item_id,size);
  ADD CONSTRAINT fk_size_itemid FOREIGN KEY (item_id) REFERENCES apparel (item_id);
```

### 4.5.2 Article_tudum Table example data

INSERT INTO article_tudum (article_title, publishing_time, category, excerpt, content_id) VALUES
('Sadie Sink on Max’s Trip to the Upside Down in ‘Stranger Things’ Season 4', '2022-06-29 11:30:00', 'meet the cast', 'Sadie Sink discusses Max\'s journey in Season 4, including her painful experiences and the significance of \"Running Up That Hill.\"', 5), ('The Duffer Brothers on Why ‘Stranger Things’ Season 4 Is Two Volumes', '2022-04-06 10:00:00', 'deep dive', 'The Duffer Brothers discuss the decision to split Season 4 into two volumes, teasing a grand finale as long as a movie.', 5),('The Queen\'s Gambit Cast, News, Videos and more', '2020-10-23 10:30:00', 'news', 'An overview of \"The Queen\'s Gambit,\" including cast information, news, and videos.', 3),('Who Is Eddie? Steve and Dustin’s ‘Stranger Things’ Friendship', '2022-06-22 09:00:00', 'meet the cast', 'Joe Keery, Gaten Matarazzo, and new star Joe Quinn preview the fourth season and their beloved bromance.', 5),(‘‘Stranger Things’ Animated Series Announced: What The Duffer Brothers Revealed So Far', '2022-07-01 13:00:00', 'news', 'Information about the new animated series set in the Stranger Things universe, in partnership with Shawn Levy\'s 21 Laps and Eric Robles.', 5),(‘‘Stranger Things’ Cast Red Carpet Premiere Photos', '2022-05-17 17:30:00', 'meet the cast', 'Photos from the red carpet premieres in Paris, Berlin, and Madrid, featuring stars like Noah Schnapp, Natalia Dyer, Charlie Heaton, and Priah Ferguson.', 5),(‘‘Stranger Things’ Season 4 Easter Eggs', '2022-06-04 12:45:00', 'explainer', 'A breakdown of Easter eggs in Season 4, including references to Kobe Bryant and Tom Cruise\'s \"Risky Business\" haircut.', 5),(‘‘Stranger Things’ Season 4 Episode Length Revealed', '2022-05-16 14:00:00', 'news', 'Details about the episode lengths for Season 4, including 7 episodes in Volume 1 and 2 episodes in Volume 2, totaling approximately 13 hours.', 5),(‘‘Stranger Things’ Season 4 First Look Photos', '2022-05-18 16:00:00', 'deep dive', 'First look photos from Season 4, showcasing the darkest and boldest chapter yet.', 5),(‘‘Stranger Things’ Season 4 Release Date Announced', '2022-02-17 08:30:00', 'news', 'The Duffer Brothers announce the release date for Season 4 and confirm that the series will conclude with Season 5.', 5);

## 4.6 Author Table

The author table represents the author of articles on Netflix Tudum.

### 4.6.1 Author Table DDL script

CREATE TABLE author (
  author_name varchar(50) NOT NULL COMMENT 'An author"s name.',

```
  description varchar(300) NOT NULL COMMENT 'The description of an author.'
) ENGINE=InnoDB DEFAULT CHARSET=utf8;
ALTER TABLE author
  ADD PRIMARY KEY (author_name);
```

### 4.6.2 Author table example data

```
INSERT INTO author (author_name, description) VALUES
('Chris Taylor', 'A journalist who covers film, TV, and celebrity culture for multiple platforms.'),
('Daniel Lee', 'An author known for writing about the evolution of TV series and their cultural impact.'),
('David Martinez', 'A film critic who writes extensively about new releases and their impact on the industry.'),
('Emily Johnson', 'A writer with a passion for in-depth analyses of popular TV series and films.'),
('James Moore', 'A seasoned writer with expertise in Netflix original content and the world of streaming.'),
('Jane Smith', 'A pop culture critic and journalist, specializing in Netflix original content.'),
('Jessica Clark', 'A journalist who specializes in casting news and trends in TV and film.'),
('John Doe', 'A seasoned entertainment writer with a focus on TV shows and movies.'),
('Laura Harris', 'A pop culture enthusiast and journalist with a focus on Netflix originals.'),
('Maria Garcia', 'An entertainment journalist with a focus on interviews and content analysis.'),
('Michael Brown', 'A long-time contributor to online entertainment publications, covering TV and film.'),
('Olivia Davis', 'A writer with a focus on behind-the-scenes details and production insights.'),
('Robert White', 'A TV and film writer known for his reviews and deep-dive content on Netflix shows.'),
('Sarah Williams', 'An entertainment writer known for her insightful reviews and interviews with actors.'),
('Sophia Anderson', 'An entertainment writer who dives deep into the stories behind popular media.');
```

## 4.7 Author_write_article Table

The author_write_article table represents a M:N “write” relationship between articles and the authors.

### 4.7.1 Author_write_article Table DDL script

```
CREATE TABLE author_write_article (
author_name varchar(50) NOT NULL COMMENT 'Author names',
 article_title varchar(100) NOT NULL COMMENT 'Article title'
) ENGINE=InnoDB DEFAULT CHARSET=utf8;
ALTER TABLE author_write_article
  ADD PRIMARY KEY (author_name,article_title),
  ADD KEY fk_write_articletitle (article_title);
  ADD CONSTRAINT fk_write_articletitle FOREIGN KEY (article_title)
REFERENCES article_tudum (article_title),
  ADD CONSTRAINT fk_write_authorname FOREIGN KEY (author_name)
REFERENCES author (author_name);
```

### 4.7.2 Author_write_article table example data

```
INSERT INTO author_write_article (author_name, article_title) VALUES
('John Doe', 'Sadie Sink on Max’s Trip to the Upside Down in ‘Stranger Things’ Season 4'),
('Maria Garcia', 'Sadie Sink on Max’s Trip to the Upside Down in ‘Stranger Things’ Season 4'),
('Jane Smith', 'The Duffer Brothers on Why ‘Stranger Things’ Season 4 Is Two Volumes'),
('John Doe', 'The Duffer Brothers on Why ‘Stranger Things’ Season 4 Is Two Volumes'),
('Chris Taylor', 'The Queen\'s Gambit Cast, News, Videos and more'),
('Sarah Williams', 'The Queen\'s Gambit Cast, News, Videos and more'),
('Emily Johnson', 'Who Is Eddie? Steve and Dustin’s ‘Stranger Things’ Friendship'),
('Michael Brown', 'Who Is Eddie? Steve and Dustin’s ‘Stranger Things’ Friendship'),
('James Moore', ''‘Stranger Things’ Animated Series Announced: What The Duffer Brothers Revealed So Far'),
('Jessica Clark', ''‘Stranger Things’ Animated Series Announced: What The Duffer Brothers Revealed So Far'),
('Daniel Lee', ''‘Stranger Things’ Cast Red Carpet Premiere Photos'),
('David Martinez', ''‘Stranger Things’ Cast Red Carpet Premiere Photos'),
('Olivia Davis', ''‘Stranger Things’ Cast Red Carpet Premiere Photos'),
```

('Sophia Anderson', ''Stranger Things' Cast Red Carpet Premiere Photos'),
('John Doe', ''Stranger Things' Season 4 Easter Eggs'),
('Laura Harris', ''Stranger Things' Season 4 Easter Eggs'),
('Maria Garcia', ''Stranger Things' Season 4 Easter Eggs'),
('Robert White', ''Stranger Things' Season 4 Easter Eggs'),
('Chris Taylor', ''Stranger Things' Season 4 Episode Length Revealed'),
('Laura Harris', ''Stranger Things' Season 4 Episode Length Revealed'),
('Robert White', ''Stranger Things' Season 4 Episode Length Revealed'),
('Sarah Williams', ''Stranger Things' Season 4 Episode Length Revealed'),
('Daniel Lee', ''Stranger Things' Season 4 First Look Photos'),
('David Martinez', ''Stranger Things' Season 4 First Look Photos'),
('Olivia Davis', ''Stranger Things' Season 4 First Look Photos'),
('Sophia Anderson', ''Stranger Things' Season 4 First Look Photos'),
('Emily Johnson', ''Stranger Things' Season 4 Release Date Announced'),
('James Moore', ''Stranger Things' Season 4 Release Date Announced'),
('Jessica Clark', ''Stranger Things' Season 4 Release Date Announced'),
('Michael Brown', ''Stranger Things' Season 4 Release Date Announced');

## 4.8 Collectible Table

The collectible is a specialization of the merchandise item table. It represents collectible items, which are items made primarily for displaying purposes, that are offered on Netflix Shop.

### 4.8.1 Collectible Table DDL script

```
CREATE TABLE collectible (
  item_id int(11) NOT NULL COMMENT 'Item ID.',
  edition int(2) NOT NULL COMMENT 'Edition number of the collection.',
  release_date date NOT NULL COMMENT 'Release date of the collection.'
) ENGINE=InnoDB DEFAULT CHARSET=utf8;
ALTER TABLE collectible
  ADD PRIMARY KEY (item_id);
  ADD CONSTRAINT fk_collectible_itemid FOREIGN KEY (item_id)
REFERENCES merchandise_item (item_id);
```

### 4.8.2 Collectible table example data

```
INSERT INTO collectible (item_id, edition, release_date) VALUES
(9, 1, '2023-04-15'),(10, 2, '2023-06-01'),(11, 1, '2023-09-20'),(12, 3,
'2023-11-10');
```

## 4.9 Content Table

The content table represents the information of shows offered on Netflix. It can be specialized into two types: ‘movie’, and ‘tv_show’.

### 4.9.1 Content Table DDL script

```
CREATE TABLE content (
  content_id int(11) NOT NULL COMMENT 'Content ID',
  content_title varchar(70) NOT NULL COMMENT 'Content title',
  release_date date NOT NULL COMMENT 'Release date',
  expiration_date date DEFAULT NULL COMMENT 'Expiration date',
  maturity_rating varchar(5) NOT NULL COMMENT 'Maturity rating',
  description text NOT NULL COMMENT 'Description'
) ENGINE=InnoDB DEFAULT CHARSET=utf8;
ALTER TABLE content
  ADD PRIMARY KEY (content_id);
```

### 4.9.2 Content table example data

```
INSERT INTO content (content_id, content_title, release_date, expiration_date, maturity_rating, description) VALUES
(1, 'A Killer Paradox', '2024-01-15', NULL, 'TV-MA', 'A 2024 thriller where a retail worker inadvertently kills a serial killer, leading to unexpected consequences.'),
(2, 'The Lincoln Lawyer', '2022-05-13', NULL, 'TV-14', 'A legal drama following a defense attorney who runs his practice from a Lincoln Town Car, navigating complex cases and personal challenges.'),
(3, 'The Queen\'s Gambit', '2020-10-23', NULL, 'TV-MA', 'A period drama about a young chess prodigy\'s rise in the male-dominated world of competitive chess during the Cold War era.'),
(4, 'One Piece', '2023-08-31', NULL, 'TV-14', 'A live-action adaptation of the popular manga and anime, following Monkey D. Luffy and his crew in search of the legendary One Piece treasure.'),
(5, 'Stranger Things', '2016-07-15', NULL, 'TV-14', 'A sci-fi horror series set in the 1980s, where a group of kids uncover supernatural mysteries in their small town.'),
(6, 'The Crown', '2016-11-04', NULL, 'TV-MA', 'A historical drama chronicling the reign of Queen Elizabeth II, exploring political and personal events that shaped the second half of the 20th century.'),
(7, 'Bridgerton', '2020-12-25', '2025-12-25', 'TV-MA', 'A romantic drama set in Regency-era London, focusing on the Bridgerton family and their entanglements in high society.'),
```

(8, 'Money Heist', '2017-05-02', NULL, 'TV-MA', 'A Spanish heist crime drama where a group of robbers, led by \'The Professor,\' execute meticulously planned heists.'),

(9, 'The Witcher', '2019-12-20', NULL, 'TV-MA', 'A fantasy series following Geralt of Rivia, a monster hunter navigating a turbulent world filled with magic and political intrigue.'),

(10, 'Squid Game', '2021-09-17', NULL, 'TV-MA', 'A South Korean survival drama where contestants in dire financial situations compete in deadly children\'s games for a massive cash prize.'),

(11, 'The Killer', '2024-10-30', NULL, 'TV-MA', 'A 2024 thriller directed by David Fincher, following an assassin who begins to question his own sanity.'),

(12, 'The Harder They Fall', '2021-10-22', NULL, 'R', 'A 2021 Western featuring an all-Black cast, where an outlaw reunites his gang to seek revenge.'),

(13, 'Red Notice', '2021-11-05', NULL, 'PG-13', 'A 2021 action-comedy starring Dwayne Johnson, Gal Gadot, and Ryan Reynolds, involving an Interpol agent tracking the world\'s most wanted art thief.'),

(14, 'Don\'t Look Up', '2021-12-24', NULL, 'R', 'A 2021 satirical science fiction film where two astronomers attempt to warn humanity about an approaching comet that will destroy Earth.'),

(15, 'The Power of the Dog', '2021-11-17', NULL, 'R', 'A 2021 Western drama exploring themes of love, grief, and resentment between two brothers on a Montana ranch.'),

(16, 'Bird Box', '2018-12-14', '2024-12-14', 'TV-MA', 'A post-apocalyptic thriller where survivors must navigate a world where seeing mysterious creatures leads to death.'),

(17, 'Marriage Story', '2019-11-06', NULL, 'R', 'A 2019 drama about a couple undergoing a painful divorce, examining their emotions and the toll it takes on their family.'),

(18, 'The Old Guard', '2020-07-10', NULL, 'R', 'A 2020 action movie about immortal warriors who fight to protect humanity, starring Charlize Theron.'),

(19, 'The Irishman', '2019-11-27', NULL, 'R', 'A 2019 epic crime film directed by Martin Scorsese, focusing on the life of a mob hitman and his involvement with organized crime.'),

(20, 'Enola Holmes', '2020-09-23', NULL, 'PG-13', 'A 2020 mystery-adventure film about Sherlock Holmes\'s younger sister, Enola, who uncovers a conspiracy while searching for her missing mother.');

### 4.10 Content_cast Table

The content_cast table represents the cast members of each show as a multivalued attribute.

### 4.10.1 Content_cast Table DDL script

```
CREATE TABLE content_cast (
  content_id int(11) NOT NULL COMMENT 'Content ID',
  cast varchar(50) NOT NULL COMMENT 'Cast name'
) ENGINE=InnoDB DEFAULT CHARSET=utf8;
ALTER TABLE content_cast
  ADD PRIMARY KEY (content_id,cast);
  ADD CONSTRAINT fk_cast_contentid FOREIGN KEY (content_id)
REFERENCES content (content_id);
```

### 4.10.2 Content_cast table example data

```
INSERT INTO content_cast (content_id, cast) VALUES
(1, 'Michael Fassbender'),(1, 'Tilda Swinton'),(2, 'Idris Elba'),
(2, 'Jonathan Majors'),(3, 'Dwayne Johnson'),(3, 'Gal Gadot'),
(3, 'Ryan Reynolds'),(4, 'Jennifer Lawrence'),(4, 'Leonardo DiCaprio'),
(4, 'Meryl Streep'),(5, 'Benedict Cumberbatch'),(5, 'Jesse Plemons'),
(5, 'Kirsten Dunst'),(6, 'John Malkovich'),(6, 'Sandra Bullock'),
(6, 'Trevante Rhodes'),(7, 'Adam Driver'),(7, 'Laura Dern'),
(7, 'Scarlett Johansson'),(8, 'Charlize Theron'),(8, 'Chiwetel Ejiofor'),
(8, 'KiKi Layne'),(9, 'Al Pacino'),(9, 'Joe Pesci'),(9, 'Robert De Niro'),
(10, 'Henry Cavill'),(10, 'Millie Bobby Brown'),(10, 'Sam Claflin'),
(11, 'Michael Fassbender'),(11, 'Tilda Swinton'),(12, 'Idris Elba'),
(12, 'Jonathan Majors'),(13, 'Dwayne Johnson'),(13, 'Gal Gadot'),
(13, 'Ryan Reynolds'),(14, 'Jennifer Lawrence'),(14, 'Leonardo DiCaprio'),
(14, 'Meryl Streep'),(15, 'Benedict Cumberbatch'),(15, 'Jesse Plemons'),
(15, 'Kirsten Dunst'),(16, 'John Malkovich'),(16, 'Sandra Bullock'),
(16, 'Trevante Rhodes'),(17, 'Adam Driver'),(17, 'Laura Dern'),
(17, 'Scarlett Johansson'),(18, 'Charlize Theron'),(18, 'Chiwetel Ejiofor'),
(18, 'KiKi Layne'),(19, 'Al Pacino'),(19, 'Joe Pesci'),(19, 'Robert De Niro'),
(20, 'Henry Cavill'),(20, 'Millie Bobby Brown'),(20, 'Sam Claflin');
```

## 4.11 Content_Director Table

The content director table represents the directors on each show as a multivalued attribute.

### 4.11.1 Content_Director DDL Script

```
CREATE TABLE content_director (
```

```
        content_id int(11) NOT NULL COMMENT 'Content ID',
 director varchar(50) NOT NULL COMMENT 'Director name'
  ) ENGINE=InnoDB DEFAULT CHARSET=utf8;
  ALTER TABLE content_director
  ADD PRIMARY KEY (content_id,director);
  ALTER TABLE content_director
  ADD CONSTRAINT fk_director_contentid FOREIGN KEY (content_id)
REFERENCES content (content_id);
```

#### 4.11.2 Content_Director table example data

```
INSERT INTO content_director (content_id, director) VALUES
(1, 'Cary Joji Fukunaga'),(1, 'David Fincher'),(2, 'Jeymes Samuel'),(3, 'Rawson
Marshall Thurber'),(4, 'Adam McKay'),
(5, 'Jane Campion'),(6, 'Susanne Bier'),(7, 'Noah Baumbach'),(8, 'Gina
Prince-Bythewood'),(9, 'Martin Scorsese'),
(10, 'Harry Bradbeer'),(11, 'David Fincher'),(12, 'Jeymes Samuel'),(13, 'Rawson
Marshall Thurber'),
(14, 'Adam McKay'),(15, 'Jane Campion'),(16, 'Susanne Bier'),(17, 'Noah
Baumbach'),(18, 'Gina Prince-Bythewood'),
(19, 'Martin Scorsese'),(20, 'Harry Bradbeer');
```

### 4.12 Content_Writer Table

The content writer table represents the writers of each show as a multivalued attribute.

#### 4.12.1 Content_Writer DDL Script

```
  CREATE TABLE content_writer (
  content_id int(11) NOT NULL COMMENT 'Content ID',
  writer varchar(50) NOT NULL COMMENT 'Writer name'
   ) ENGINE=InnoDB DEFAULT CHARSET=utf8;
ALTER TABLE content_writer
  ADD PRIMARY KEY (content_id,writer);
ALTER TABLE content_writer
  ADD CONSTRAINT fk_writer_contentid FOREIGN KEY (content_id)
REFERENCES content (content_id);
```

#### 4.12.2 Content_Writer table example data

```
INSERT INTO content_writer (content_id, writer) VALUES
(1, 'Andrew Kevin Walker'), (1, 'David Fincher'), (2, 'Jeymes Samuel'),(3, 'Rawson
Marshall Thurber'),(4, 'Adam McKay'), (4, 'David Sirota'), (5, 'Jane Campion'), (6,
'Susanne Bier'), (7, 'Noah Baumbach'), (8, 'Gina Prince-Bythewood'),
 (8, 'Greg Rucka'), (9, 'Jay Cocks'),(9, 'Steven Zaillian'),(10, 'Jack Thorne'),(11,
'David Fincher'),(12, 'Jeymes Samuel'),
(13, 'Rawson Marshall Thurber'),(14, 'Adam McKay'),(14, 'David Sirota'),(15,
'Jane Campion'),(16, 'Susanne Bier'),
(17, 'Noah Baumbach'),(18, 'Gina Prince-Bythewood'),(18, 'Greg Rucka'),
(19, 'Jay Cocks'),(19, 'Steven Zaillian'),(20, 'Jack Thorne');
```

## 4.13 Coupon Table

The coupon table represents the coupon used

### 4.13.1 Coupon DDL Script

```
CREATE TABLE coupon (
  promo_code varchar(10) NOT NULL COMMENT 'Promotion code.',
  expiration_date date NOT NULL COMMENT 'Coupon expiration date.',
  discount_percent int(3) NOT NULL COMMENT 'Discount percent offered by
coupon.',
  is_valid_for_all_items tinyint(1) NOT NULL COMMENT 'Mark if coupon is valid
for all items.'
) ENGINE=InnoDB DEFAULT CHARSET=utf8;
ALTER TABLE coupon
  ADD PRIMARY KEY (promo_code);
```

### 4.13.2 Coupon table example data

```
INSERT INTO coupon (promo_code, expiration_date, discount_percent,
is_valid_for_all_items) VALUES
('BLACKFRI', '2024-11-29', 25, 1),('BRIDGE15', '2024-11-30', 15, 0),('FALLDEAL',
'2024-11-30', 10, 1),
('NEWYEAR5', '2024-01-01', 5, 1),('SPRING25', '2024-03-31', 25, 1),('SQUID20',
'2024-10-31', 20, 0),
('STRANGE10', '2024-12-31', 10, 0),('SUMMER10', '2024-06-30', 10,
1),('WITCHER5', '2024-12-31', 5, 0),
('XMAS2024', '2024-12-25', 30, 1);
```

## 4.14 Coupon_eligiblefor_item Table

The coupon_eligiblefor_item holds information on which merchandise item is eligible for which coupon on Netflix Shop.

### 4.14.1 coupon_eligiblefor_item DDL Script

```
CREATE TABLE coupon_eligiblefor_item (
  promo_code varchar(10) NOT NULL COMMENT 'Coupon"s promo code',
  item_id int(20) NOT NULL COMMENT 'Merchandise item ID.'
) ENGINE=InnoDB DEFAULT CHARSET=utf8;
ALTER TABLE coupon_eligiblefor_item
  ADD PRIMARY KEY (promo_code,item_id),
  ADD KEY fk_validfor_itemid (item_id);
  ADD CONSTRAINT fk_validfor_itemid FOREIGN KEY (item_id) REFERENCES
merchandise_item (item_id),
  ADD CONSTRAINT fk_validfor_promocode FOREIGN KEY (promo_code)
REFERENCES coupon (promo_code);
```

### 4.14.2 coupon_eligiblefor_item table example data

```
INSERT INTO coupon_eligiblefor_item (promo_code, item_id) VALUES
('STRANGE10', 1),('WITCHER5', 2),('BRIDGE15', 3),('SQUID20', 4),
('STRANGE10', 5),('WITCHER5', 6),('BRIDGE15', 7),('SQUID20', 8),
('STRANGE10', 9),('WITCHER5', 10),('BRIDGE15', 11),('SQUID20', 12);
```

## 4.15 Drinkware Table

The drinkware table is a specialization of the merchandise item. It represents the drinkware items offered on Netflix Shop, such as water bottles and cups.

### 4.15.1 Drinkware DDL Script

```
CREATE TABLE drinkware (
  item_id int(11) NOT NULL,
  capacity int(5) NOT NULL COMMENT 'Capacity of drinkware, measured in ml.',
  upper_diameter int(2) NOT NULL COMMENT 'Upper diameter of drinkware,
measured in inches.',
  bottom_diameter int(2) NOT NULL COMMENT 'Bottom diameter of drinkware,
measured in inches.'
) ENGINE=InnoDB DEFAULT CHARSET=utf8;
ALTER TABLE drinkware
```

```
  ADD PRIMARY KEY (item_id);
ALTER TABLE drinkware
  ADD CONSTRAINT fk_drinkware_itemid FOREIGN KEY (item_id)
REFERENCES merchandise_item (item_id);
```

#### 4.15.2 DrinkWare table example data

```
INSERT INTO drinkware (item_id, capacity, upper_diameter, bottom_diameter)
VALUES
(5, 350, 3, 3),(6, 450, 3, 2),(7, 500, 3, 3),(8, 750, 3, 2);
```

### 4.16 Game Table

The game table represents games offered on the Netflix Games program.

#### 4.16.1 Game DDL Script

```
CREATE TABLE game (
  game_name varchar(50) NOT NULL COMMENT 'A game"s name',
  description text NOT NULL COMMENT 'Game"s description',
  internet_requirement_status tinyint(1) NOT NULL COMMENT 'Is internet
required to play or not?',
  maturity_rating varchar(4) NOT NULL COMMENT 'Maturity rating',
  player_count varchar(5) NOT NULL COMMENT 'Min/max amount of player
required to play the game.',
  release_year year(4) NOT NULL COMMENT 'Release year',
  controller_support_status tinyint(1) NOT NULL COMMENT 'Is controller
supported or not?',
  compatibility_details text NOT NULL COMMENT 'Short description of the device
specs needed to play this game.'
) ENGINE=InnoDB DEFAULT CHARSET=utf8;
```

#### 4.16.2 Game table example data

```
INSERT INTO game (game_name, description, internet_requirement_status,
maturity_rating, player_count, release_year, controller_support_status,
compatibility_details) VALUES
('Asphalt Xtreme', 'Off-road racing game with various vehicle classes and
multiplayer.', 1, 'E10+', '1-8', 2021, 1, 'Needs stable internet and devices with
good graphics support.'),
('Exploding Kittens', 'Card game adaptation with humorous strategy elements.', 0,
'E', '2-5', 2021, 0, 'Requires Android 5.0 or iOS 11 and up.'),
```

('Heads Up!', 'A fun party game where players guess words based on clues.', 1, 'E', '2+', 2018, 0, 'Requires iOS or Android with camera access.'),
('Hextech Mayhem', 'Rhythm-based runner set in the League of Legends universe.', 1, 'T', '1', 2021, 1, 'Compatible with most modern devices with touchscreen.'), ('Into the Breach', 'Turn-based strategy where players control mechs to defend cities from aliens.', 0, 'E10+', '1', 2018, 1, 'Compatible with mid-range devices, iOS and Android.'),

('Kentucky Route Zero', 'A magical realist adventure game exploring a mysterious highway.', 0, 'T', '1', 2019, 1, 'Requires Android 6.0 or iOS 12 and up.'),
('Moonlighter', 'Action RPG with dungeon crawling and shop management.', 0, 'T', '1', 2018, 1, 'Runs on most Android 6.0+ and iOS 12+ devices.'),
('Oxenfree', 'Supernatural thriller adventure game involving dialogue choices.', 0, 'T', '1', 2016, 1, 'Requires Android 5.0 or iOS 11 and up.'),
('Shatter Remastered', 'A modernized version of a brick-breaking game with power-ups.', 0, 'E', '1', 2022, 1, 'Compatible with iOS and Android devices from 2018 onward.'),
('Stranger Things: 1984', 'Pixel-art action game set in the Stranger Things universe.', 0, 'T', '1', 2020, 1, 'Playable on low-end Android and iOS devices.');

## 4.17 Game_Category Table

The game category table represents how games on Netflix may belong to multiple categories, thus being represented by a multivalued attribute table.

### 4.17.1 Game_Category Table DDL Script

```
CREATE TABLE game_category (
  game_name varchar(50) NOT NULL COMMENT 'Game name',
  category varchar(20) NOT NULL COMMENT 'Game Category'
) ENGINE=InnoDB DEFAULT CHARSET=utf8;
ALTER TABLE game_category
  ADD PRIMARY KEY (category,game_name),
  ADD KEY fk_mtval_gamecategory (game_name);
```

### 4.17.2 Game_Category Table Example Data

```
INSERT INTO game_category (game_name, category) VALUES
('Asphalt Xtreme', 'Multiplayer'),('Asphalt Xtreme', 'Racing'),
('Exploding Kittens', 'Card Game'),('Exploding Kittens', 'Casual'),
```

```
('Heads Up!', 'Party Game'),('Heads Up!', 'Trivia'),
('Hextech Mayhem', 'Action'),('Hextech Mayhem', 'Rhythm'),
('Into the Breach', 'Strategy'),('Into the Breach', 'Turn-Based'),
('Kentucky Route Zero', 'Adventure'),('Kentucky Route Zero', 'Narrative'),
('Moonlighter', 'Action RPG'),('Moonlighter', 'Dungeon Crawler'),
('Oxenfree', 'Adventure'),('Oxenfree', 'Thriller'),
('Shatter Remastered', 'Arcade'),('Shatter Remastered', 'Puzzle'),
('Stranger Things: 1984', 'Action'),('Stranger Things: 1984', 'Retro');
```

## 4.18 Game_Developer Table

The game developer table represents the developers on a game as a multivalued attribute.

### 4.18.1 Game_Developer Table DDL Script

```
CREATE TABLE game_developer (
  game_name varchar(50) NOT NULL COMMENT 'Game name',
  developer varchar(50) NOT NULL COMMENT 'The game''s developer'
) ENGINE=InnoDB DEFAULT CHARSET=utf8;
ALTER TABLE game_developer
  ADD PRIMARY KEY (developer,game_name),
  ADD KEY fk_mtval_gamedev (game_name);
ALTER TABLE game_developer
  ADD CONSTRAINT fk_mtval_gamedev FOREIGN KEY (game_name)
REFERENCES game (game_name);
```

### 4.18.2 Game_Developer Table Example Data

```
INSERT INTO game_developer (game_name, developer) VALUES
('Asphalt Xtreme', 'Gameloft'),('Exploding Kittens', 'Exploding Kittens, Inc.'),
('Heads Up!', 'Warner Bros. Games'),('Hextech Mayhem', 'Choice Provisions'),
('Into the Breach', 'Subset Games'),('Kentucky Route Zero', 'Cardboard
Computer'),
('Moonlighter', 'Digital Sun'),('Oxenfree', 'Night School Studio'),
('Shatter Remastered', 'PikPok'),('Stranger Things: 1984', 'BonusXP');
```

## 4.19 Game_Gamemode Table

The game_gamemode table represents the different game modes for games on Netflix as a multivalued attribute table.

### 4.19.1 Game_Gamemode Table DDL Script

```
CREATE TABLE game_gamemode (
  game_name varchar(50) NOT NULL COMMENT 'Game name',
  game_mode varchar(20) NOT NULL COMMENT 'The different game modes of
the game.'
) ENGINE=InnoDB DEFAULT CHARSET=utf8;
ALTER TABLE game_gamemode
  ADD PRIMARY KEY (game_name,game_mode);
ALTER TABLE game_gamemode
  ADD CONSTRAINT fk_mtval_gamemode FOREIGN KEY (game_name)
REFERENCES game (game_name);
```

### 4.19.2 Game_Gamemode Table Example Data

```
INSERT INTO game_gamemode (game_name, game_mode) VALUES
('Asphalt Xtreme', 'Multiplayer'),('Asphalt Xtreme', 'Single Player'),
('Exploding Kittens', 'Multiplayer'),('Exploding Kittens', 'Single Player'),
('Heads Up!', 'Multiplayer'),('Hextech Mayhem', 'Single Player'),
('Into the Breach', 'Single Player'),('Kentucky Route Zero', 'Single Player'),
('Moonlighter', 'Single Player'),('Oxenfree', 'Single Player'),
('Shatter Remastered', 'Single Player'),('Stranger Things: 1984', 'Single Player');
```

## 4.20 Game_Supported_Languages Table

The game_supported_languages table represents the multiple languages a game may support as a multivalued attribute table.

### 4.20.1 Game_Supported_Languages Table DDL Script

```
CREATE TABLE game_supported_languages (
  game_name varchar(50) NOT NULL COMMENT 'Game name',
        supported_lang varchar(20) NOT NULL COMMENT 'What languages the
game supports.'
        ) ENGINE=InnoDB DEFAULT CHARSET=utf8;
ALTER TABLE game_supported_languages
  ADD PRIMARY KEY (game_name,supported_lang);
        ALTER TABLE game_supported_languages
  ADD CONSTRAINT fk_suplang_gamename FOREIGN KEY (game_name)
REFERENCES game (game_name);
```

### 4.20.2 Game_Supported_Languages Table Example Data

```
INSERT INTO game_supported_languages (game_name, supported_lang)
VALUES
('Asphalt Xtreme', 'English'),('Asphalt Xtreme', 'French'),('Asphalt Xtreme',
'Italian'),
('Exploding Kittens', 'English'),('Exploding Kittens', 'French'),('Exploding Kittens',
'Spanish'),
('Heads Up!', 'English'),('Heads Up!', 'German'),('Heads Up!', 'Spanish'),
('Hextech Mayhem', 'English'),('Hextech Mayhem', 'German'),('Hextech Mayhem',
'Portuguese'),
('Into the Breach', 'English'),('Into the Breach', 'French'),('Into the Breach',
'German'),
('Into the Breach', 'Spanish'),('Kentucky Route Zero', 'English'),('Kentucky Route
Zero', 'French'),
('Kentucky Route Zero', 'Spanish'),('Moonlighter', 'English'),('Moonlighter',
'French'),
('Moonlighter', 'German'),('Oxenfree', 'English'),('Oxenfree', 'French'),
('Oxenfree', 'German'),('Shatter Remastered', 'English'),('Shatter Remastered',
'Spanish'),
('Stranger Things: 1984', 'English'),('Stranger Things: 1984', 'Spanish');
```

### 4.21 Genre Table

The genre table represents the different genres a show on netflix may belong to. It includes a unique genre ID, and a genre name.

#### 4.21.1 Genre DDL Script

```
CREATE TABLE genre (
  genre_id int(11) NOT NULL COMMENT 'Genre ID',
  genre_name varchar(30) NOT NULL COMMENT 'Genre Name'
) ENGINE=InnoDB DEFAULT CHARSET=utf8;
ALTER TABLE genre
  ADD PRIMARY KEY (genre_id);
```

#### 4.21.2 Genre table example data

```
INSERT INTO genre (genre_id, genre_name) VALUES
(1, 'Action'),(2, 'Adventure'),(3, 'Comedy'),(4, 'Crime'),
(5, 'Drama'),(6, 'Fantasy'),(7, 'Historical'),(8, 'Horror'),
```

(9, 'Mystery'),(10, 'Romance'),(11, 'Science Fiction'),
(12, 'Thriller'),(13, 'Western');

### 4.22 Genre_Contain_Content Table

The genre_contain_content table represents the M:N “contain” relationship between the genres and shows on Netflix.

#### 4.22.1 Genre_Contain_Content DDL Script

```
CREATE TABLE genre_contain_content (
  genre_id int(11) NOT NULL COMMENT 'Genre ID',
  content_id int(11) NOT NULL COMMENT 'Content ID'
) ENGINE=InnoDB DEFAULT CHARSET=utf8;
ALTER TABLE genre_contain_content
  ADD PRIMARY KEY (genre_id,content_id),
  ADD KEY fk_contain_contentid (content_id);
ALTER TABLE genre_contain_content
  ADD CONSTRAINT fk_contain_contentid FOREIGN KEY (content_id)
REFERENCES content (content_id),
  ADD CONSTRAINT fk_contain_genreid FOREIGN KEY (genre_id)
REFERENCES genre (genre_id);
```

#### 4.22.2 Genre_Contain_Content Example Data

```
INSERT INTO genre_contain_content (genre_id, content_id) VALUES
(4, 1),(12, 1),(5, 2),(13, 2),(1, 3),(2, 3),(3, 3),(3, 4),(5, 4),(5, 5),(10, 5),(5, 6),(12,
6),(3, 7),(5, 7),(1, 8),(2, 8),(4, 9),
(5, 9),(2, 10),(9, 10),(5, 11),(6, 11),(5, 12),(6, 12),(2, 13),(5, 13),(8, 13),(4, 14),(5,
14),(12, 14),(4, 15),(5, 15),
(5, 16),(10, 16),(5, 17),(9, 17),(12, 17),(5, 18),(7, 18),(1, 19),(2, 19),(5, 19),(4,
20),(5, 20),(12, 20);
```

### 4.23 merchandise_item table

The merchandise_item table represents all items offered on the Netflix Shop. It can be specialized into three different types: ‘apparel’, ‘drinkware’, and ‘collectible’.

#### 4.23.1 merchandise_item table DDL Script

```
CREATE TABLE merchandise_item (
  item_id int(11) NOT NULL COMMENT 'Item ID',
  collection_category varchar(50) NOT NULL COMMENT 'The collection category this item belongs to.',
  price decimal(10,2) NOT NULL COMMENT 'Price',
  description varchar(300) NOT NULL COMMENT 'Description',
  item_name varchar(50) NOT NULL COMMENT 'Item Name',
  discounted_price decimal(10,2) DEFAULT NULL COMMENT 'Discounted price.',
  stock_quantity int(11) NOT NULL COMMENT 'Number of items remaining in stock.'
) ENGINE=InnoDB DEFAULT CHARSET=utf8;

ALTER TABLE merchandise_item
  ADD PRIMARY KEY (item_id);
```

#### 4.23.2 merchandise_item Example Data

```
INSERT INTO merchandise_item (item_id, collection_category, price, description, item_name, discounted_price, stock_quantity) VALUES
(1, 'Stranger Things Collection', '29.99', 'Stranger Things Logo T-Shirt', 'Stranger Things T-Shirt', NULL, 100),
(2, 'The Witcher Collection', '39.99', 'The Witcher Pullover Hoodie', 'The Witcher Hoodie', NULL, 50),
(3, 'Bridgerton Collection', '25.99', 'Bridgerton Lace Tee', 'Bridgerton T-Shirt', NULL, 75),
(4, 'Squid Game Collection', '49.99', 'Squid Game Player Tracksuit', 'Squid Game Tracksuit', '44.99', 30),
(5, 'Stranger Things Collection', '15.99', 'Stranger Things themed mug with high-quality ceramic material.', 'Stranger Things Mug', '12.99', 50),
(6, 'Bridgerton Collection', '19.99', 'Bridgerton crystal wine glass set, ideal for any occasion.', 'Bridgerton Wine Glass', NULL, 30),
(7, 'The Witcher Collection', '12.99', 'The Witcher inspired stainless steel tumbler for hot or cold drinks.', 'The Witcher Tumbler', NULL, 40),
(8, 'Squid Game Collection', '25.99', 'Squid Game glass bottle with iconic symbols from the show.', 'Squid Game Glass Bottle', '22.99', 20),
(9, 'Stranger Things Collection', '14.99', 'Limited edition Stranger Things Funko Pop featuring Eleven.', 'Stranger Things Funko Pop', NULL, 100),
(10, 'The Witcher Collection', '9.99', 'Detailed keychain inspired by The Witcher series, featuring Geralt\'s medallion.', 'The Witcher Keychain', NULL, 200),
```

(11, 'Squid Game Collection', '19.99', 'Exclusive Squid Game figure with collectible display stand.', 'Squid Game Figure', NULL, 150),
(12, 'Bridgerton Collection', '24.99', 'Elegant Bridgerton mini replica of the Queen\'s crown for display.', 'Bridgerton Crown Replica', NULL, 75);

### 4.24 merchandise_order Table

The merchandise order table represents an order made by a user on the Netflix Shop.

#### 4.24.1 merchandise_order DDL Script

```
CREATE TABLE merchandise_order (
  order_id int(11) NOT NULL COMMENT 'Order"s ID',
  username varchar(20) NOT NULL COMMENT 'User name',
  contact_email varchar(254) NOT NULL COMMENT 'Contact Email',
  card_number varchar(19) NOT NULL COMMENT 'Card Number on a credit card.',
  expiration_date char(5) NOT NULL COMMENT 'Expiration Date on credit card.',
  security_code char(3) NOT NULL COMMENT 'Security code on a credit card.',
  name_on_card varchar(26) NOT NULL COMMENT 'Name on the credit card.',
  fname varchar(50) NOT NULL COMMENT 'First Name',
  lname varchar(50) NOT NULL COMMENT 'Last Name',
  postal_code varchar(10) NOT NULL COMMENT 'Postal Code',
  province varchar(20) NOT NULL COMMENT 'Province',
  city varchar(20) NOT NULL COMMENT 'City',
  country varchar(20) NOT NULL COMMENT 'Country',
  addr varchar(200) DEFAULT NULL COMMENT 'Extra address information',
  order_date date NOT NULL COMMENT 'Creation date of the order.',
  status varchar(10) NOT NULL COMMENT 'Order completion status. (Shipping, Cancelled, Completed)'
) ENGINE=InnoDB DEFAULT CHARSET=utf8;
ALTER TABLE merchandise_order
  ADD PRIMARY KEY (order_id),
  ADD KEY fk_order_username (username);
ALTER TABLE merchandise_order
  ADD CONSTRAINT fk_order_username FOREIGN KEY (username)
REFERENCES user_account (username);
```

#### 4.24.2 merchandise_order example data

```
INSERT INTO merchandise_order (order_id, username, contact_email,
card_number, expiration_date, security_code, name_on_card, fname, lname,
postal_code, province, city, country, addr, order_date, status) VALUES
(1, 'alice_b', 'alice.b@example.com', '4532 1234 5678 9012', '02/34', '321', 'Alice
Brown', 'Alice', 'Brown', '12345', 'Ontario', 'Toronto', 'Canada', '901 Chestnut St.',
'2024-10-01', 'completed'),
(2, 'bob_the_builder', 'bob.builder@example.com', '4716 2345 6789 0123',
'01/35', '654', 'Bob Builder', 'Bob', 'Builder', '67890', 'Quebec', 'Montreal',
'Canada', '902 Cedar Ave.', '2024-10-03', 'shipped'),
(3, 'charlie123', 'charlie123@example.com', '4916 3456 7890 1234', '12/36', '987',
'Charlie Lee', 'Charlie', 'Lee', '13579', 'British Columbia', 'Vancouver', 'Canada',
'903 Maple Rd.', '2024-10-05', 'cancelled'),
(4, 'david_k', 'david.k@example.com', '4532 4567 8901 2345', '11/37', '210',
'David King', 'David', 'King', '24680', 'Alberta', 'Calgary', 'Canada', '904 Pine
Cres.', '2024-10-07', 'completed'),
(5, 'emily.r', 'emily.r@example.com', '4716 5678 9012 3456', '10/38', '543', 'Emily
Rose', 'Emily', 'Rose', '11223', 'Manitoba', 'Winnipeg', 'Canada', '905 Spruce Ln.',
'2024-10-09', 'shipped'),
(6, 'frank88', 'frank88@example.com', '4916 6789 0123 4567', '09/39', '876',
'Frank Gold', 'Frank', 'Gold', '44556', 'Nova Scotia', 'Halifax', 'Canada', '906 Birch
Blvd.', '2024-10-11', 'completed'),
(7, 'gina_l', 'gina.l@example.com', '4532 7890 1234 5678', '08/40', '109', 'Gina
Lake', 'Gina', 'Lake', '88990', 'Ontario', 'Ottawa', 'Canada', '907 Oak Pl.',
'2024-10-13', 'cancelled'),
(8, 'harry_p', 'harry.p@example.com', '4716 8901 2345 6789', '07/41', '432',
'Harry Potter', 'Harry', 'Potter', '22334', 'Newfoundland', 'St. John\'s', 'Canada',
'908 Aspen Ct.', '2024-10-15', 'shipped'),
(9, 'jane_smith', 'jane.smith@example.com', '4916 9012 3456 7890', '06/42',
'765', 'Jane Smith', 'Jane', 'Smith', '55678', 'New Brunswick', 'Moncton', 'Canada',
'909 Redwood Dr.', '2024-10-17', 'completed'),
(10, 'john_doe', 'john.doe@example.com', '4532 0123 4567 8901', '05/43', '098',
'John Doe', 'John', 'Doe', '78901', 'Yukon', 'Whitehorse', 'Canada', '910 Alder
Way', '2024-10-19', 'completed'),
(11, 'alice_b', 'alice.b@example.com', '4532 1234 5678 9012', '02/34', '321', 'Alice
Brown', 'Alice', 'Brown', '12345', 'Ontario', 'Toronto', 'Canada', '901 Chestnut St.',
'2024-11-01', 'shipped'),
(12, 'bob_the_builder', 'bob.builder@example.com', '4716 2345 6789 0123',
'01/35', '654', 'Bob Builder', 'Bob', 'Builder', '67890', 'Quebec', 'Montreal',
'Canada', '902 Cedar Ave.', '2024-11-03', 'completed'),
```

(13, 'charlie123', 'charlie123@example.com', '4916 3456 7890 1234', '12/36', '987', 'Charlie Lee', 'Charlie', 'Lee', '13579', 'British Columbia', 'Vancouver', 'Canada', '903 Maple Rd.', '2024-11-05', 'shipped'),
(14, 'david_k', 'david.k@example.com', '4532 4567 8901 2345', '11/37', '210', 'David King', 'David', 'King', '24680', 'Alberta', 'Calgary', 'Canada', '904 Pine Cres.', '2024-11-07', 'cancelled'),
(15, 'emily.r', 'emily.r@example.com', '4716 5678 9012 3456', '10/38', '543', 'Emily Rose', 'Emily', 'Rose', '11223', 'Manitoba', 'Winnipeg', 'Canada', '905 Spruce Ln.', '2024-11-09', 'completed'),
(16, 'frank88', 'frank88@example.com', '4916 6789 0123 4567', '09/39', '876', 'Frank Gold', 'Frank', 'Gold', '44556', 'Nova Scotia', 'Halifax', 'Canada', '906 Birch Blvd.', '2024-11-11', 'shipped'),
(17, 'gina_l', 'gina.l@example.com', '4532 7890 1234 5678', '08/40', '109', 'Gina Lake', 'Gina', 'Lake', '88990', 'Ontario', 'Ottawa', 'Canada', '907 Oak Pl.', '2024-11-13', 'completed'),
(18, 'harry_p', 'harry.p@example.com', '4716 8901 2345 6789', '07/41', '432', 'Harry Potter', 'Harry', 'Potter', '22334', 'Newfoundland', 'St. John\'s', 'Canada', '908 Aspen Ct.', '2024-11-15', 'completed'),
(19, 'jane_smith', 'jane.smith@example.com', '4916 9012 3456 7890', '06/42', '765', 'Jane Smith', 'Jane', 'Smith', '55678', 'New Brunswick', 'Moncton', 'Canada', '909 Redwood Dr.', '2024-11-17', 'cancelled'),
(20, 'john_doe', 'john.doe@example.com', '4532 0123 4567 8901', '05/43', '098', 'John Doe', 'John', 'Doe', '78901', 'Yukon', 'Whitehorse', 'Canada', '910 Alder Way', '2024-11-19', 'shipped'),
(21, 'alice_b', 'alice.b@example.com', '4532 1234 5678 9012', '02/34', '321', 'Alice Brown', 'Alice', 'Brown', '12345', 'Ontario', 'Toronto', 'Canada', '901 Chestnut St.', '2024-10-03', 'shipped'),
(22, 'bob_the_builder', 'bob.builder@example.com', '4716 2345 6789 0123', '01/35', '654', 'Bob Builder', 'Bob', 'Builder', '67890', 'Quebec', 'Montreal', 'Canada', '902 Cedar Ave.', '2024-10-06', 'completed'),
(23, 'charlie123', 'charlie123@example.com', '4916 3456 7890 1234', '12/36', '987', 'Charlie Lee', 'Charlie', 'Lee', '13579', 'British Columbia', 'Vancouver', 'Canada', '903 Maple Rd.', '2024-10-09', 'cancelled'),
(24, 'david_k', 'david.k@example.com', '4532 4567 8901 2345', '11/37', '210', 'David King', 'David', 'King', '24680', 'Alberta', 'Calgary', 'Canada', '904 Pine Cres.', '2024-10-12', 'completed'),
(25, 'emily.r', 'emily.r@example.com', '4716 5678 9012 3456', '10/38', '543', 'Emily Rose', 'Emily', 'Rose', '11223', 'Manitoba', 'Winnipeg', 'Canada', '905 Spruce Ln.', '2024-10-15', 'shipped'),

(26, 'frank88', 'frank88@example.com', '4916 6789 0123 4567', '09/39', '876', 'Frank Gold', 'Frank', 'Gold', '44556', 'Nova Scotia', 'Halifax', 'Canada', '906 Birch Blvd.', '2024-10-18', 'completed'),
(27, 'gina_l', 'gina.l@example.com', '4532 7890 1234 5678', '08/40', '109', 'Gina Lake', 'Gina', 'Lake', '88990', 'Ontario', 'Ottawa', 'Canada', '907 Oak Pl.', '2024-10-21', 'cancelled'),
(28, 'harry_p', 'harry.p@example.com', '4716 8901 2345 6789', '07/41', '432', 'Harry Potter', 'Harry', 'Potter', '22334', 'Newfoundland', 'St. John\'s', 'Canada', '908 Aspen Ct.', '2024-10-24', 'shipped'),
(29, 'jane_smith', 'jane.smith@example.com', '4916 9012 3456 7890', '06/42', '765', 'Jane Smith', 'Jane', 'Smith', '55678', 'New Brunswick', 'Moncton', 'Canada', '909 Redwood Dr.', '2024-10-27', 'completed'),
(30, 'john_doe', 'john.doe@example.com', '4532 0123 4567 8901', '05/43', '098', 'John Doe', 'John', 'Doe', '78901', 'Yukon', 'Whitehorse', 'Canada', '910 Alder Way', '2024-10-30', 'completed');

### 4.25 merchandise_review Table

The merchandise_review table represents the reviews made by users, for items on Netflix Shop.

#### 4.24.1 merchandise_review  DDL Script

```
CREATE TABLE merchandise_review (
  review_id int(11) NOT NULL COMMENT 'Review ID',
  username varchar(20) NOT NULL COMMENT 'User name',
  item_id int(11) NOT NULL COMMENT 'Item ID',
  review_title varchar(50) NOT NULL COMMENT 'Review Title',
  comment varchar(200) NOT NULL COMMENT 'Textual content of the review.',
  rating int(1) NOT NULL COMMENT 'Rating given.',
  name varchar(20) NOT NULL COMMENT 'Name displayed on the review.',
  creation_date date NOT NULL COMMENT 'Creation date.',
  email varchar(254) NOT NULL COMMENT 'Email',
  verified_buyer_flag tinyint(1) NOT NULL COMMENT 'Mark if user is a verified buyer of this item.'
) ENGINE=InnoDB DEFAULT CHARSET=utf8;
ALTER TABLE merchandise_review
  ADD PRIMARY KEY (review_id,username,item_id),
  ADD KEY fk_review_username (username),
  ADD KEY fk_review_itemid (item_id)
ALTER TABLE merchandise_review
```

```
  ADD CONSTRAINT fk_review_itemid FOREIGN KEY (item_id) REFERENCES
merchandise_item (item_id),
  ADD CONSTRAINT fk_review_username FOREIGN KEY (username)
REFERENCES user_account (username);
```

### 4.24.2 merchandise_review example data

```
INSERT INTO merchandise_review (review_id, username, item_id, review_title,
comment, rating, name, creation_date, email, verified_buyer_flag) VALUES
(1, 'john_doe', 1, 'Perfect Fit!', 'The Stranger Things T-Shirt fits great and feels
comfortable. The design is awesome!', 5, 'John D.', '2024-10-12',
'john.doe@example.com', 1),
(2, 'jane_smith', 2, 'Cozy and Warm', 'The Witcher Hoodie is perfect for chilly
days. Love the design!', 5, 'Jane S.', '2024-10-13', 'jane.smith@example.com', 1),
(3, 'alice_b', 3, 'Not as Expected', 'The Bridgerton T-Shirt didn\'t meet my
expectations. The lace feels cheap.', 2, 'Alice B.', '2024-10-14',
'alice.b@example.com', 0),
(4, 'bob_the_builder', 4, 'Great Value', 'Got the Squid Game Tracksuit on sale. It
fits well and looks exactly like in the show!', 4, 'Bob B.', '2024-10-15',
'bob.builder@example.com', 1),
(5, 'charlie123', 5, 'Nice Mug', 'Stranger Things Mug is nice, but I wish it were a
bit bigger.', 4, 'Charlie', '2024-10-16', 'charlie123@example.com', 1),
(6, 'david_k', 6, 'Elegant Glasses', 'The Bridgerton Wine Glasses are stunning.
Perfect for entertaining!', 5, 'David K.', '2024-10-17', 'david.k@example.com', 1),
(7, 'emily.r', 7, 'Works Well', 'The Witcher Tumbler is great for hot coffee. Keeps it
warm for hours.', 5, 'Emily R.', '2024-10-18', 'emily.r@example.com', 0),
(8, 'frank88', 8, 'Stylish Bottle', 'The Squid Game Glass Bottle looks amazing, but
it\'s a bit fragile.', 3, 'Frank88', '2024-10-19', 'frank88@example.com', 1),
(9, 'gina_l', 9, 'Love It!', 'The Stranger Things Funko Pop of Eleven is so cute! A
must-have for fans.', 5, 'Gina L.', '2024-10-20', 'gina.l@example.com', 1),
(10, 'harry_p', 10, 'Not Worth It', 'The Witcher Keychain is a bit overpriced for
what it is. Small but nice design.', 3, 'Harry P.', '2024-10-21',
'harry.p@example.com', 0),
(11, 'john_doe', 11, 'Amazing Detail', 'The Squid Game Figure looks great!
Detailed and worth collecting.', 5, 'John D.', '2024-10-22',
'john.doe@example.com', 1),
(12, 'jane_smith', 12, 'Beautiful Display Piece', 'The Bridgerton Crown Replica is
exquisite. Looks amazing on my shelf.', 5, 'Jane S.', '2024-10-23',
'jane.smith@example.com', 1),
(13, 'alice_b', 1, 'Nice Tee', 'The Stranger Things T-Shirt is cool, but the fabric
could be softer.', 4, 'Alice B.', '2024-10-24', 'alice.b@example.com', 1),
```

(14, 'bob_the_builder', 4, 'Very Comfortable', 'The Squid Game Tracksuit is super comfortable. The fit is true to size.', 5, 'Bob B.', '2024-10-25', 'bob.builder@example.com', 1),
(15, 'charlie123', 5, 'Good Quality', 'Stranger Things Mug is well made, perfect for a hot drink.', 5, 'Charlie', '2024-10-26', 'charlie123@example.com', 1),
(16, 'david_k', 8, 'Unique Design', 'Love the Squid Game Glass Bottle. The design is unique and stands out.', 4, 'David K.', '2024-10-27', 'david.k@example.com', 1),
(17, 'emily.r', 9, 'Collector\'s Item', 'Stranger Things Funko Pop is amazing for collectors. Highly recommended.', 5, 'Emily R.', '2024-10-28', 'emily.r@example.com', 1),
(18, 'frank88', 7, 'Functional and Cool', 'The Witcher Tumbler works well and looks great. Perfect for fans.', 5, 'Frank88', '2024-10-29', 'frank88@example.com', 0),
(19, 'gina_l', 2, 'Warm and Cozy', 'The Witcher Hoodie is very comfortable and warm. I wear it all the time.', 5, 'Gina L.', '2024-10-30', 'gina.l@example.com', 1),
(20, 'harry_p', 10, 'Disappointing', 'The Witcher Keychain is smaller than expected and feels cheap.', 2, 'Harry P.', '2024-10-31', 'harry.p@example.com', 0),
(21, 'john_doe', 3, 'Just OK', 'The Bridgerton T-Shirt is just alright. Expected a bit more quality.', 3, 'John D.', '2024-11-01', 'john.doe@example.com', 1),
(22, 'jane_smith', 12, 'Stunning', 'The Bridgerton Crown Replica is beautiful. Looks very regal on my shelf.', 5, 'Jane S.', '2024-11-02', 'jane.smith@example.com', 1),
(23, 'alice_b', 6, 'Elegant Set', 'The Bridgerton Wine Glasses look fantastic. Very elegant design.', 5, 'Alice B.', '2024-11-03', 'alice.b@example.com', 0),
(24, 'bob_the_builder', 8, 'Great Design', 'Squid Game Glass Bottle is stylish but a bit fragile. Handle with care.', 4, 'Bob B.', '2024-11-04', 'bob.builder@example.com', 1),
(25, 'charlie123', 9, 'Awesome Figure', 'The Stranger Things Funko Pop is a great addition to my collection!', 5, 'Charlie', '2024-11-05', 'charlie123@example.com', 1),
(26, 'david_k', 11, 'Worth It!', 'The Squid Game Figure is amazing. Great quality and detail.', 5, 'David K.', '2024-11-06', 'david.k@example.com', 1),
(27, 'emily.r', 4, 'Soft Fabric', 'The Squid Game Tracksuit is super soft and comfy. I love it!', 5, 'Emily R.', '2024-11-07', 'emily.r@example.com', 1),
(28, 'frank88', 1, 'Not Bad', 'Stranger Things T-Shirt is nice but runs a bit small. Order a size up!', 3, 'Frank88', '2024-11-08', 'frank88@example.com', 0),
(29, 'gina_l', 7, 'Sturdy Tumbler', 'The Witcher Tumbler is very sturdy and keeps drinks hot for hours.', 5, 'Gina L.', '2024-11-09', 'gina.l@example.com', 1),

(30, 'harry_p', 5, 'Cool Mug!', 'The Stranger Things Mug looks great and is very durable.', 4, 'Harry P.', '2024-11-10', 'harry.p@example.com', 1);

### 4.26 movie Table

The movie table represents the movie type of show on Netflix. It is a specialization of the content table.

#### 4.25.1 movie DDL Script

```
CREATE TABLE movie (
  content_id int(11) NOT NULL COMMENT 'Content ID',
  movieduration time NOT NULL COMMENT 'Movie duration'
) ENGINE=InnoDB DEFAULT CHARSET=utf8;
ALTER TABLE movie
  ADD PRIMARY KEY (content_id);
ALTER TABLE movie
  ADD CONSTRAINT fk_movie_contentid FOREIGN KEY (content_id)
REFERENCES content (content_id);
```

#### 4.25.2 movie example data

```
INSERT INTO movie (content_id, movieduration) VALUES
(11, '02:01:00'),(12, '02:19:00'),(13, '01:58:00'),(14, '02:18:00'),(15,
'02:06:00'),(16, '02:04:00'),(17, '02:16:00'),
(18, '02:05:00'),(19, '03:29:00'),(20, '02:03:00');
```

### 4.27 order_contains_item Table

The order_contains_item table represents the M:N “contain” relationship between orders and merchandise items on Netflix Shop.

#### 4.27.1 order_contains_item DDL Script

```
CREATE TABLE order_contains_item (
  order_id int(11) NOT NULL COMMENT 'The ID of the order',
  item_id int(11) NOT NULL COMMENT 'The ID of the merchandise item',
  quantity int(11) NOT NULL COMMENT 'The quantity of the item in the order'
) ENGINE=InnoDB DEFAULT CHARSET=utf8;
ALTER TABLE order_contains_item
  ADD PRIMARY KEY (order_id,item_id),
  ADD KEY item_id (item_id);
ALTER TABLE order_contains_item
```

```
  ADD CONSTRAINT order_contains_item_ibfk_2 FOREIGN KEY (item_id)
REFERENCES merchandise_item (item_id),
  ADD CONSTRAINT order_contains_item_ibfk_3 FOREIGN KEY (order_id)
REFERENCES merchandise_order (order_id);
```

#### 4.27.2 order_contains_item example data

```
INSERT INTO order_contains_item (order_id, item_id, quantity) VALUES
(1, 1, 2),(1, 3, 1),(2, 2, 3),(2, 4, 1),(3, 5, 2),(3, 6, 1),(4, 7, 3),(5, 8, 2),(5, 9, 1),(6,
10, 4),(7, 11, 3),(7, 12, 1),(8, 1, 2),
(9, 2, 1),(9, 3, 2),(10, 4, 3),(10, 5, 1),(11, 6, 4),(11, 7, 1),(12, 8, 2),(13, 9, 3),(13,
10, 1),(14, 11, 2),(15, 12, 4),(16, 1, 3),
(16, 5, 1),(17, 2, 4),(17, 6, 2),(18, 7, 1),(18, 8, 3),(19, 9, 2),(19, 10, 4),(20, 11,
3),(21, 12, 2),(22, 1, 1),(22, 3, 3),
(23, 4, 2),(23, 5, 4),(24, 6, 3),(25, 7, 2),(25, 8, 1),(26, 9, 4),(27, 10, 2),(28, 11,
1),(29, 12, 3),(30, 1, 4);
```

### 4.28 order_use_coupon Table

The order_use_coupon table represents the M:N “use” relationship between coupons and orders.

#### 4.28.1 order_use_coupon DDL Script

```
CREATE TABLE order_use_coupon (
  order_id int(11) NOT NULL COMMENT 'The ID of the order',
  promo_code varchar(10) NOT NULL COMMENT 'The coupon code used in the
order'
) ENGINE=InnoDB DEFAULT CHARSET=utf8;
ALTER TABLE order_use_coupon
  ADD PRIMARY KEY (order_id,promo_code),
  ADD UNIQUE KEY unique_order_coupon (order_id,promo_code),
  ADD KEY promo_code (promo_code);
ALTER TABLE order_use_coupon
  ADD CONSTRAINT order_use_coupon_ibfk_1 FOREIGN KEY (order_id)
REFERENCES merchandise_order (order_id),
  ADD CONSTRAINT order_use_coupon_ibfk_2 FOREIGN KEY (promo_code)
REFERENCES coupon (promo_code);
```

#### 4.28.2 order_use_coupon example data

```
INSERT INTO order_use_coupon (order_id, promo_code) VALUES
(6, 'BLACKFRI'),(15, 'BLACKFRI'),(1, 'BRIDGE15'),(9, 'BRIDGE15'),(11,
'BRIDGE15'),(5, 'FALLDEAL'),
```

(12, 'FALLDEAL'),(14, 'NEWYEAR5'),(22, 'NEWYEAR5'),(23, 'NEWYEAR5'),(30, 'SPRING25'),(2, 'SQUID20'),
\(12, 'SQUID20'),(1, 'STRANGE10'),(13, 'STRANGE10'),(16, 'STRANGE10'),(4, 'SUMMER10'),(9, 'SUMMER10'),
(17, 'SUMMER10'),(19, 'SUMMER10'),(25, 'SUMMER10'),(26, 'SUMMER10'),(2, 'WITCHER5'),(6, 'WITCHER5'),
(9, 'WITCHER5'),(11, 'WITCHER5'),(13, 'WITCHER5'),(2, 'XMAS2024'),(10, 'XMAS2024'),(20, 'XMAS2024');

## 4.29 payment_method Table

The payment_method table represents a payment method a user account can create, in order to subscribe to one of Netflix plans.

### 4.29.1 payment_method DDL Script

```
CREATE TABLE payment_method (
  card_number varchar(19) NOT NULL COMMENT 'Credit card number',
  username varchar(20) NOT NULL COMMENT 'The username of the user owning this payment payment method.',
  expiration_date char(5) NOT NULL COMMENT 'Expiration date of credit card',
  security_code char(3) NOT NULL COMMENT 'Security code of credit card.',
  name_on_card varchar(26) NOT NULL COMMENT 'Name on credit card.'
) ENGINE=InnoDB DEFAULT CHARSET=utf8;
ALTER TABLE payment_method
  ADD PRIMARY KEY (card_number),
  ADD KEY fk_paymentmethod_username (username);
ALTER TABLE payment_method
  ADD CONSTRAINT fk_paymentmethod_username FOREIGN KEY (username) REFERENCES user_account (username);
```

### 4.29.2 payment_method example date

```
INSERT INTO payment_method (card_number, username, expiration_date, security_code, name_on_card) VALUES
('3714 4963 0399 505', 'emily.r', '09/23', '345', 'Emily Roberts'),
('3787 3449 3676 001', 'emily.r', '08/25', '234', 'Emily Roberts'),
('4111 1111 1111 1111', 'alice_b', '12/25', '123', 'Alice Brown'),
('4111 1111 1111 1117', 'jane_smith', '01/25', '678', 'Jane Smith'),
('4111 1111 1111 1120', 'david_k', '09/26', '012', 'David Knight'),
('4111 1111 1111 1130', 'charlie123', '03/27', '678', 'Charlie Brown'),
('4500 0000 0000 0011', 'bob_the_builder', '11/26', '345', 'Bob Builder'),
('5105 1051 0510 5100', 'harry_p', '03/25', '345', 'Harry Park'),
('5222 2222 2222 2224', 'david_k', '12/24', '012', 'David Knight'),
```

('5500 0000 0000 0004', 'bob_the_builder', '06/24', '456', 'Bob Builder'),
('5500 0000 0000 0005', 'frank88', '04/24', '567', 'Frank Green'),
('6011 0000 0000 0009', 'charlie123', '10/23', '789', 'Charlie Brown'),
('6011 0000 0000 0010', 'gina_l', '02/27', '890', 'Gina Lee'),
('6011 0000 0000 0022', 'alice_b', '05/23', '234', 'Alice Brown'),
('6200 0000 0000 0003', 'john_doe', '07/24', '901', 'John Doe');

## 4.30 plan Table

The plan table represents the different subscription plans offered by Netflix.

### 4.30.1 plan DDL Script

```
CREATE TABLE plan (
  plan_name char(10) NOT NULL COMMENT 'Plan name',
  monthly_price decimal(4,2) NOT NULL COMMENT 'Monthly price',
  vid_sound_quality varchar(15) NOT NULL COMMENT 'Video and sound
quality',
  resolution varchar(10) NOT NULL COMMENT 'Content resolution',
  ads_enabled_flag tinyint(1) NOT NULL COMMENT 'Mark if plan is ads
enabled.',
  simul_watch_count int(1) NOT NULL COMMENT 'Count how many devices are
allowed to watch on this account.',
  download_device_count int(1) NOT NULL COMMENT 'Count how many
devices are allowed to download content using this account.',
  extra_member_slots int(1) NOT NULL COMMENT 'Count how many other user
accounts you can invite to become extra members.'
) ENGINE=InnoDB DEFAULT CHARSET=utf8;
ALTER TABLE plan
  ADD PRIMARY KEY (plan_name);
```

### 4.30.2 plan example data

```
INSERT INTO plan (plan_name, monthly_price, vid_sound_quality, resolution,
ads_enabled_flag, simul_watch_count, download_device_count,
extra_member_slots) VALUES
('BASIC', '9.99', 'Stereo', 'SD', 0, 1, 1, 0),
('BASICADS', '6.99', 'Stereo', 'SD', 1, 1, 1, 0),
('MOBILE', '3.99', 'Stereo', 'SD', 0, 1, 1, 0),
('PREMIUM', '19.99', 'Dolby Atmos', 'UHD/4K', 0, 4, 6, 2),
('STANDARD', '15.99', 'Dolby Audio', 'HD', 0, 2, 2, 1);
```

### 4.31 Profile Table

The profile table represents a profile that belongs to a user account. Each use account has a minimum of 1 profile, and can have up to 5 profiles.

#### 4.31.1 Profile Table DDL Script

```
CREATE TABLE profile (
  profile_name varchar(20) NOT NULL COMMENT 'Profile name',
  username varchar(20) NOT NULL COMMENT 'User name',
  maturity_rating varchar(3) NOT NULL COMMENT 'Maturity rating',
  kids_profile_status tinyint(1) NOT NULL COMMENT 'Mark if the profile is a
"kids profile".',
  display_language varchar(3) NOT NULL COMMENT 'UI display language',
  audio_sub_language varchar(10) NOT NULL COMMENT 'Audio and subtitles
language',
  eps_autoplay_flag tinyint(1) NOT NULL COMMENT 'Mark if next episode in a
series should play automatically after the last one ends.',
  preview_autoplay_flag tinyint(1) NOT NULL COMMENT 'Mark if a preview for a
content should play automatically when hovering.',
  pin int(6) DEFAULT NULL COMMENT 'PIN to secure profile access.',
  game_handle varchar(20) DEFAULT NULL COMMENT 'Game handle for user
profile playing Netflix Games.'
) ENGINE=InnoDB DEFAULT CHARSET=utf8;
ALTER TABLE profile
  ADD PRIMARY KEY (profile_name,username),
  ADD KEY fk_profile_username (username);
ALTER TABLE profile
  ADD CONSTRAINT fk_profile_username FOREIGN KEY (username)
REFERENCES user_account (username);
```

#### 4.31.2 Profile table example data

```
INSERT INTO profile (profile_name, username, maturity_rating,
kids_profile_status, display_language, audio_sub_language, eps_autoplay_flag,
preview_autoplay_flag, pin, game_handle) VALUES
('AdventureZone', 'bob_the_builder', '18+', 0, 'ENG', 'ENG', 1, 1, 654321,
'bob149'),
('BeastMode', 'frank88', '18+', 0, 'ENG', 'ENG', 1, 1, 223344, 'frank257'),
('CaptainElite', 'harry_p', '18+', 0, 'ENG', 'ENG', 0, 1, 778899, 'harry259'),
('CosmicQueen', 'charlie123', '18+', 0, 'ENG', 'FRE', 0, 1, 112233, 'charlie223'),
```

```
('LilBuddy', 'bob_the_builder', 'G', 1, 'ENG', 'GER', 1, 0, NULL, NULL),
('LilDreamer', 'emily.r', 'G', 1, 'ENG', 'POR', 1, 0, NULL, NULL),
('Maverick', 'emily.r', '18+', 0, 'ENG', 'ENG', 0, 1, 998877, 'emily321'),
('MiniExplorer', 'charlie123', 'G', 1, 'ENG', 'ENG', 1, 0, NULL, NULL),
('RogueMaster', 'gina_l', '18+', 0, 'ENG', 'ENG', 1, 1, 445566, 'gina512'),
('SpaceCadet', 'david_k', 'G', 1, 'ENG', 'ITA', 1, 0, NULL, NULL),
('SuperHeroX', 'alice_b', '18+', 0, 'ENG', 'ENG', 1, 1, 123456, 'alice257'),
('TechGuru', 'david_k', '18+', 0, 'ENG', 'ENG', 1, 1, 334455, 'david789'),
('TinyAdventurer', 'frank88', 'G', 1, 'ENG', 'SPA', 1, 0, NULL, NULL),
('TinyTots', 'alice_b', 'G', 1, 'ENG', 'SPA', 1, 0, NULL, NULL),
('WonderKid', 'gina_l', 'G', 1, 'ENG', 'ENG', 1, 0, NULL, NULL);
```

### 4.32 Subscription Table

The subscription table represents the ternary relationship “subscribe” between user_account, “plan”, and “payment_method”.

#### 4.32.1 Subscription DDL Script

```
CREATE TABLE subscription (
  subscription_id int(11) NOT NULL COMMENT 'Subscription ID.',
  plan_name char(10) NOT NULL COMMENT 'Plan name',
  username varchar(20) NOT NULL COMMENT 'User name',
  card_number varchar(19) NOT NULL COMMENT 'Card number',
  subscription_date date NOT NULL COMMENT 'Subscription date',
  termination_date date DEFAULT NULL COMMENT 'Termination date. (Null if
subscription is still active)'
) ENGINE=InnoDB DEFAULT CHARSET=utf8;
ALTER TABLE subscription
  ADD PRIMARY KEY (subscription_id),
  ADD KEY fk_subscription_cardnumber (card_number),
  ADD KEY fk_subscrption_username (username),
  ADD KEY fk_subscription_planname (plan_name);
ALTER TABLE subscription
  ADD CONSTRAINT fk_subscription_cardnumber FOREIGN KEY
(card_number) REFERENCES payment_method (card_number),
  ADD CONSTRAINT fk_subscription_planname FOREIGN KEY (plan_name)
REFERENCES plan (plan_name),
  ADD CONSTRAINT fk_subscrption_username FOREIGN KEY (username)
REFERENCES user_account (username);
```

### 4.32.2 Subscription table example data

INSERT INTO subscription (subscription_id, plan_name, username, card_number, subscription_date, termination_date) VALUES (1, 'BASIC', 'alice_b', '4111 1111 1111 1111', '2024-01-01', '2024-06-01'),(2, 'BASICADS', 'alice_b', '6011 0000 0000 0022', '2024-06-02', NULL),(3, 'MOBILE', 'bob_the_builder', '4500 0000 0000 0011', '2023-03-01', '2024-05-01'),(4, 'PREMIUM', 'bob_the_builder', '5500 0000 0000 0004', '2024-05-02', NULL),(5, 'BASIC', 'charlie123', '4111 1111 1111 1130', '2024-02-01', '2024-04-01'),(6, 'BASICADS', 'charlie123', '6011 0000 0000 0009', '2024-04-02', NULL),(7, 'MOBILE', 'david_k', '4111 1111 1111 1120', '2023-11-01', '2024-02-01'),(8, 'PREMIUM', 'david_k', '5222 2222 2222 2224', '2024-02-02', NULL),(9, 'BASIC', 'emily.r', '3714 4963 0399 505', '2023-10-01', '2024-03-01'),(10, 'STANDARD', 'emily.r', '3787 3449 3676 001', '2024-03-02', NULL),(11, 'BASIC', 'frank88', '5500 0000 0000 0005', '2023-09-01', '2024-02-01'),(12, 'MOBILE', 'frank88', '6011 0000 0000 0010', '2024-02-02', NULL),(13, 'STANDARD', 'gina_l', '6011 0000 0000 0010', '2023-12-01', '2024-05-01'),(14, 'PREMIUM', 'gina_l', '5500 0000 0000 0005', '2024-05-02', NULL),(15, 'BASIC', 'harry_p', '5105 1051 0510 5100', '2023-08-01', '2024-02-01'),(16, 'STANDARD', 'harry_p', '4111 1111 1111 1117', '2024-02-02', NULL),(17, 'PREMIUM', 'jane_smith', '6200 0000 0000 0003', '2023-07-01', '2024-01-01'),(18, 'BASICADS', 'jane_smith', '4111 1111 1111 1117', '2024-01-02', NULL),(19, 'STANDARD', 'john_doe', '6011 0000 0000 0022', '2024-02-01', '2024-04-01'),(20, 'PREMIUM', 'john_doe', '4111 1111 1111 1111', '2024-04-02', NULL),(21, 'BASIC', 'alice_b', '6011 0000 0000 0022', '2024-06-01', '2024-08-01'),(22, 'PREMIUM', 'alice_b', '4500 0000 0000 0011', '2024-08-02', NULL),(23, 'MOBILE', 'bob_the_builder', '5500 0000 0000 0004', '2024-07-01', '2024-09-01'),(24, 'BASICADS', 'bob_the_builder', '4500 0000 0000 0011', '2024-09-02', NULL),(25, 'STANDARD', 'charlie123', '6011 0000 0000 0009', '2024-05-01', '2024-07-01'),(26, 'PREMIUM', 'charlie123', '5222 2222 2222 2224', '2024-07-02', NULL),(27, 'BASIC', 'david_k', '4111 1111 1111 1120', '2024-04-01', '2024-06-01'),(28, 'PREMIUM', 'david_k', '5222 2222 2222 2224', '2024-06-02', NULL),(29, 'STANDARD', 'emily.r', '3714 4963 0399 505', '2024-01-01', '2024-03-01'),(30, 'BASICADS', 'emily.r', '3787 3449 3676 001', '2024-03-02', NULL);

## 4.33 tv_episode Table

The tv_episode table represents an episode contained within a TV show on Netflix.

### 4.33.1 tv_episode DDL Script

```
CREATE TABLE tv_episode (
  content_id int(11) NOT NULL COMMENT 'Content ID of the episode.',
  season_no int(11) NOT NULL COMMENT 'The number of season the episode
belongs to.',
  ep_no int(11) NOT NULL COMMENT 'The number of the episode.',
  ep_name varchar(100) NOT NULL COMMENT 'An episode''s name',
  description varchar(300) NOT NULL COMMENT 'The episode''s description',
  episode_duration int(11) NOT NULL COMMENT 'An episode''s duration in
seconds.'
) ENGINE=InnoDB DEFAULT CHARSET=utf8;
ALTER TABLE tv_episode
  ADD PRIMARY KEY (content_id,season_no,ep_no);
ALTER TABLE tv_episode
  ADD CONSTRAINT tv_episode_ibfk_1 FOREIGN KEY (content_id)
REFERENCES tv_show (content_id);
```

### 4.33.2 tv_episode  table example data

```
INSERT INTO tv_episode (content_id, season_no, ep_no, ep_name, description,
episode_duration) VALUES
(1, 1, 1, 'Episode 1: The Awakening', 'The protagonist begins a life-changing
investigation into a paradox.', 2730),
(1, 1, 2, 'Episode 2: Paradox Revealed', 'The investigation takes a dramatic turn
as new information surfaces.', 2760),
(1, 1, 3, 'Episode 3: The Final Answer', 'All the mysteries come to a head in this
thrilling conclusion.', 2830),
(2, 1, 1, 'Episode 1: First Case', 'The lawyer takes on his first case that seems
impossible to win.', 2910),
(2, 1, 2, 'Episode 2: Legal Troubles', 'The lawyer faces difficulties as his client’s
past comes back to haunt them.', 2940),
(2, 1, 3, 'Episode 3: The Courtroom Battle', 'The lawyer takes the case to court,
where things get intense.', 3000),
(3, 1, 1, 'Episode 1: The Beginning', 'Beth Harmon discovers her passion for
chess at an early age.', 3000),
(3, 1, 2, 'Episode 2: The Challenge', 'Beth faces her first big competition, a
challenging tournament.', 3120),
```

(3, 1, 3, 'Episode 3: The Rival', 'Beth faces off against her toughest opponent yet in the chess world.', 3090),
(4, 1, 1, 'Episode 1: The Pirate King\'s Will', 'Luffy begins his journey to become the next Pirate King.', 2700),
(4, 1, 2, 'Episode 2: The Straw Hat Crew', 'Luffy gathers his first companions for the adventure ahead.', 2790),
(4, 1, 3, 'Episode 3: The First Battle', 'The crew faces their first major battle with rival pirates.', 2820),
(4, 2, 1, 'Episode 1: The Grand Line', 'The crew sets sail for the Grand Line in search of treasure.', 2820),
(4, 2, 2, 'Episode 2: The Lost Island', 'Luffy and the crew discover an island filled with mysteries.', 2880),
(4, 2, 3, 'Episode 3: The Conqueror', 'A confrontation with a pirate king shakes the crew to their core.', 2850),
(5, 1, 1, 'Episode 1: The Vanishing', 'A young boy mysteriously disappears in Hawkins, Indiana.', 2740),
(5, 1, 2, 'Episode 2: The Upside Down', 'The group of friends discover a strange world beyond our own.', 2880),
(5, 1, 3, 'Episode 3: The Monster', 'The kids face off against a terrifying creature from the Upside Down.', 3030),
(5, 2, 1, 'Episode 1: The Return', 'The gang reunites after some time apart, but danger lurks once again.', 2800),
(5, 2, 2, 'Episode 2: The Terrors of Hawkins', 'More secrets of Hawkins are revealed as the group digs deeper into the mysteries.', 2940),
(5, 2, 3, 'Episode 3: The Final Showdown', 'The group confronts their fears in a climactic battle.', 3000),
(6, 1, 1, 'Episode 1: The New Queen', 'Elizabeth II ascends to the throne after her father\'s sudden death.', 3120),
(6, 1, 2, 'Episode 2: The Burden of Rule', 'Queen Elizabeth struggles with the weight of responsibility.', 3180),
(6, 1, 3, 'Episode 3: The First Crisis', 'A political crisis forces the new Queen to make difficult decisions.', 3240),
(6, 2, 1, 'Episode 1: The Suez Crisis', 'The Queen faces her first international crisis involving the Suez Canal.', 3270),
(6, 2, 2, 'Episode 2: The King\'s Dilemma', 'The Queen\'s husband, Prince Philip, faces his own struggles.', 3330),
(6, 2, 3, 'Episode 3: The Family Dilemma', 'The Queen must navigate familial challenges as well as political ones.', 3380),
(7, 1, 1, 'Episode 1: The Duke and the Duchess', 'The love story between Daphne and Simon begins.', 2760),

(7, 1, 2, 'Episode 2: The Ball', 'A grand ball brings the high society of London together.', 2850),
(7, 1, 3, 'Episode 3: The Secret', 'Daphne discovers a secret about Simon’s past that threatens their relationship.', 2900),
(7, 2, 1, 'Episode 1: The Proposal', 'A shocking proposal changes the course of Daphne and Simon’s future.', 2890),
(7, 2, 2, 'Episode 2: The Return', 'Daphne and Simon deal with the fallout from their decisions.', 2980),
(7, 2, 3, 'Episode 3: The Resolution', 'The couple faces their final challenges, culminating in a heartwarming conclusion.', 3030),
(8, 1, 1, 'Episode 1: The Plan', 'The professor unveils his elaborate plan to rob the Royal Mint of Spain.', 2700),
(8, 1, 2, 'Episode 2: The Robbery', 'The heist begins, and everything quickly escalates.', 2850),
(8, 1, 3, 'Episode 3: The First Obstacle', 'The robbers face their first major challenge during the heist.', 2800),
(8, 2, 1, 'Episode 1: The Escape', 'The crew begins planning their escape after the robbery goes sideways.', 2930),
(8, 2, 2, 'Episode 2: The Betrayal', 'One of the crew members betrays the group, putting everything at risk.', 2950),
(8, 2, 3, 'Episode 3: The Tension', 'Tension builds as the heist reaches its breaking point.', 3000),
(9, 1, 1, 'Episode 1: The Last Wish', 'Geralt of Rivia embarks on his first adventure involving a dangerous wish.', 2700),
(9, 1, 2, 'Episode 2: The Hunt', 'Geralt hunts down a deadly creature that has been terrorizing a village.', 2780),
(9, 1, 3, 'Episode 3: The Witcher\'s Code', 'Geralt faces a moral dilemma regarding the Witcher’s code.', 2900),
(9, 2, 1, 'Episode 1: The Warlock', 'Geralt encounters a powerful warlock who has secrets of his own.', 2820),
(9, 2, 2, 'Episode 2: The Sorceress', 'Geralt forms an unlikely alliance with a sorceress to defeat a greater evil.', 2890),
(9, 2, 3, 'Episode 3: The Betrayal', 'Geralt must make a choice that could alter his fate forever.', 2980),
(10, 1, 1, 'Episode 1: The Game Begins', 'Participants enter the deadly game with their lives at stake.', 2670),
(10, 1, 2, 'Episode 2: The First Challenge', 'The first challenge begins, and the participants realize the stakes.', 2750),
(10, 1, 3, 'Episode 3: The Betrayal', 'A shocking betrayal occurs during the second round of the game.', 2850),

(10, 2, 1, 'Episode 1: The Survivor', 'The few remaining players must face the final game of survival.', 2900),
(10, 2, 2, 'Episode 2: The Winner\'s Dilemma', 'The winner must deal with the consequences of their actions.', 2980),
(10, 2, 3, 'Episode 3: The Price of Victory', 'The final episode reveals the shocking price of victory in the Squid Game.', 3000);

### 4.34 tv_show Table

The tv_show table represents a tv_show type of show on Netflix. It is a specialization of the content table.

#### 4.34.1 tv_show DDL Script

```
CREATE TABLE tv_show (
  content_id int(11) NOT NULL COMMENT 'Content ID',
  series_status varchar(20) NOT NULL COMMENT 'The current status of the series (e.g., Ongoing, Completed, Canceled)',
  episode_release_frequency varchar(20) DEFAULT NULL COMMENT 'Frequency of episode releases (e.g., Weekly, All at Once)'
) ENGINE=InnoDB DEFAULT CHARSET=utf8;
ALTER TABLE tv_show
  ADD PRIMARY KEY (content_id);
ALTER TABLE tv_show
  ADD CONSTRAINT fk_tvshow_contentid FOREIGN KEY (content_id) REFERENCES content (content_id);
```

#### 4.34.2 tv_show  table example data

```
INSERT INTO tv_show (content_id, series_status, episode_release_frequency) VALUES
(1, 'Ongoing', 'Weekly'),(2, 'Ongoing', 'All at Once'),(3, 'Completed', 'All at Once'),(4, 'Ongoing', 'All at Once'),(5, 'Ongoing', 'All at Once'),(6, 'Ongoing', 'All at Once'),(7, 'Ongoing', 'All at Once'),(8, 'Completed', 'All at Once'),(9, 'Ongoing', 'All at Once'),(10, 'Ongoing', 'All at Once');
```

### 4.35 user_account Table

The user_account table represents the user accounts registered on netflix.

### 4.35.1 user_account DDL Script

```
CREATE TABLE user_account (
  username varchar(20) NOT NULL COMMENT 'User name',
  email varchar(254) NOT NULL COMMENT 'User Email',
  password int(64) NOT NULL COMMENT 'User password',
  date_of_birth date DEFAULT NULL COMMENT 'User date of birth',
  gender varchar(1) DEFAULT NULL COMMENT 'User gender'
) ENGINE=InnoDB DEFAULT CHARSET=utf8;
ALTER TABLE user_account
  ADD PRIMARY KEY (username);
```

### 4.35.2 user_account table example data

```
INSERT INTO user_account (username, email, password, date_of_birth, gender)
VALUES
('alice_b', 'alice.b@example.com', 34567890, NULL, NULL),('bob_the_builder',
'bob.builder@example.com', 45678901, '2000-05-12', 'M'),('charlie123',
'charlie123@example.com', 56789012, NULL, NULL),('david_k',
'david.k@example.com', 67890123, NULL, NULL),('emily.r', 'emily.r@example.com',
78901234, '1995-12-25', 'F'),('frank88', 'frank88@example.com', 89012345, NULL,
NULL),('gina_l', 'gina.l@example.com', 90123456, '1992-03-08', 'F'),('harry_p',
'harry.p@example.com', 12345679, NULL, NULL),('jane_smith',
'jane.smith@example.com', 23456789, '1985-07-21', 'F'),('john_doe',
'john.doe@example.com', 12345678, '1990-01-15', 'M');
```

## 4.36 user_download_game Table

The user_download_game table represents the M:N relationship “download” between users and games offered on the Netflix Game program.

### 4.36.1 user_download_game DDL Script

```
CREATE TABLE user_download_game (
  download_id int(11) NOT NULL COMMENT 'Download ID',
  game_name varchar(50) NOT NULL COMMENT 'Game name',
  username varchar(20) NOT NULL COMMENT 'User name',
  download_date date NOT NULL COMMENT 'The date the game is downloaded
by the user.'
) ENGINE=InnoDB DEFAULT CHARSET=utf8;
ALTER TABLE user_download_game
```

```
  ADD PRIMARY KEY (download_id),
  ADD KEY username (username),
  ADD KEY game_name (game_name);
ALTER TABLE user_download_game
  MODIFY download_id int(11) NOT NULL AUTO_INCREMENT COMMENT
'Download ID', AUTO_INCREMENT=32;
ALTER TABLE user_download_game
  ADD CONSTRAINT user_download_game_ibfk_1 FOREIGN KEY (username)
REFERENCES user_account (username),
  ADD CONSTRAINT user_download_game_ibfk_2 FOREIGN KEY
(game_name) REFERENCES game (game_name);
```

#### 4.36.2 user_download_game table example data

```
INSERT INTO user_download_game (download_id, game_name, username,
download_date) VALUES
(1, 'Asphalt Xtreme', 'alice_b', '2024-01-10'),(2, 'Shatter Remastered', 'alice_b',
'2024-11-28'),(3, 'Stranger Things: 1984', 'alice_b', '2024-12-30'),(4, 'Asphalt
Xtreme', 'bob_the_builder', '2024-11-11'),(5, 'Exploding Kittens', 'bob_the_builder',
'2024-02-12'),(6, 'Stranger Things: 1984', 'bob_the_builder', '2024-11-30'),(7,
'Asphalt Xtreme', 'charlie123', '2024-11-05'),(8, 'Exploding Kittens', 'charlie123',
'2024-12-01'),(9, 'Heads Up!', 'charlie123', '2024-03-05'),(10, 'Exploding Kittens',
'david_k', '2024-11-10'),(11, 'Heads Up!', 'david_k', '2024-12-05'),(12, 'Hextech
Mayhem', 'david_k', '2024-04-01'),(13, 'Heads Up!', 'emily.r', '2024-11-12'),(14,
'Hextech Mayhem', 'emily.r', '2024-12-12'),(15, 'Into the Breach', 'emily.r',
'2024-05-15'),(16, 'Hextech Mayhem', 'frank88', '2024-11-15'),(17, 'Into the
Breach', 'frank88', '2024-12-15'),(18, 'Kentucky Route Zero', 'frank88',
'2024-06-20'),(19, 'Into the Breach', 'gina_l', '2024-11-18'),(20, 'Kentucky Route
Zero', 'gina_l', '2024-12-20'),(21, 'Moonlighter', 'gina_l', '2024-07-17'),(22,
'Kentucky Route Zero', 'harry_p', '2024-11-20'),(23, 'Moonlighter', 'harry_p',
'2024-12-22'),(24, 'Oxenfree', 'harry_p', '2024-08-22'),(25, 'Heads Up!',
'jane_smith', '2024-12-03'),(26, 'Moonlighter', 'jane_smith', '2024-11-22'),(27,
'Oxenfree', 'jane_smith', '2024-12-25'),(28, 'Shatter Remastered', 'jane_smith',
'2024-09-09'),(29, 'Oxenfree', 'john_doe', '2024-11-25'),(30, 'Shatter Remastered',
'john_doe', '2024-12-28'),(31, 'Stranger Things: 1984', 'john_doe', '2024-10-30');
```

### 4.37 user_shown_ads Table

The user_shown_ads table logs when a user with an ad-enabled subscription plan is shown ads.

### 4.37.1 user_shown_ads DDL Script

```
CREATE TABLE user_shown_ads (
  showads_id int(11) NOT NULL COMMENT 'Showing Ads ID',
  ads_title varchar(20) NOT NULL COMMENT 'The title of an ad.',
  user_name varchar(20) NOT NULL COMMENT 'User name',
  show_datetime datetime NOT NULL COMMENT 'The datetime when this ad
was shown to the user.',
  is_skipped tinyint(1) NOT NULL COMMENT 'Mark if the ad was skipped.'
) ENGINE=InnoDB DEFAULT CHARSET=utf8;
ALTER TABLE user_shown_ads
  ADD PRIMARY KEY (showads_id),
  ADD KEY user_name (user_name),
  ADD KEY ads_title (ads_title);
ALTER TABLE user_shown_ads
  ADD CONSTRAINT user_shown_ads_ibfk_1 FOREIGN KEY (user_name)
REFERENCES user_account (username),
  ADD CONSTRAINT user_shown_ads_ibfk_2 FOREIGN KEY (ads_title)
REFERENCES ads (ads_title);
```

### 4.37.2 user_shown_ads table example data

```
INSERT INTO user_shown_ads (showads_id, ads_title, user_name,
show_datetime, is_skipped) VALUES
(1, 'Electric Cars', 'alice_b', '2024-11-01 08:30:00', 1),(2, 'Gadget Promotion',
'alice_b', '2024-11-01 09:00:00', 1),(3, 'Healthy Snacks', 'alice_b', '2024-11-01
09:30:00', 1),(4, 'Holiday Travel', 'alice_b', '2024-11-01 10:00:00', 0),(5, 'New
Movie Release', 'alice_b', '2024-11-01 10:30:00', 1),(6, 'Streaming Deals',
'alice_b', '2024-11-01 11:00:00', 1),(7, 'Electric Cars', 'charlie123', '2024-11-02
08:30:00', 1),(8, 'Gadget Promotion', 'charlie123', '2024-11-02 09:00:00', 0),(9,
'Healthy Snacks', 'charlie123', '2024-11-02 09:30:00', 1),(10, 'Holiday Travel',
'charlie123', '2024-11-02 10:00:00', 1),(11, 'New Movie Release', 'charlie123',
'2024-11-02 10:30:00', 1),(12, 'Streaming Deals', 'charlie123', '2024-11-02
11:00:00', 1),(13, 'Electric Cars', 'jane_smith', '2024-11-03 08:30:00', 1),(14,
'Gadget Promotion', 'jane_smith', '2024-11-03 09:00:00', 1),(15, 'Healthy Snacks',
'jane_smith', '2024-11-03 09:30:00', 0),(16, 'Holiday Travel', 'jane_smith',
'2024-11-03 10:00:00', 1),(17, 'New Movie Release', 'jane_smith', '2024-11-03
10:30:00', 1),(18, 'Streaming Deals', 'jane_smith', '2024-11-03 11:00:00', 1),(19,
'Electric Cars', 'bob_the_builder', '2024-11-04 08:30:00', 1),(20, 'Gadget
Promotion', 'bob_the_builder', '2024-11-04 09:00:00', 1),(21, 'Healthy Snacks',
```

'bob_the_builder', '2024-11-04 09:30:00', 1),(22, 'Holiday Travel', 'bob_the_builder', '2024-11-04 10:00:00', 0),(23, 'New Movie Release', 'bob_the_builder', '2024-11-04 10:30:00', 1),(24, 'Streaming Deals', 'bob_the_builder', '2024-11-04 11:00:00', 1),(25, 'Electric Cars', 'emily.r', '2024-11-05 08:30:00', 1),(26, 'Gadget Promotion', 'emily.r', '2024-11-05 09:00:00', 1),(27, 'Healthy Snacks', 'emily.r', '2024-11-05 09:30:00', 1),(28, 'Holiday Travel', 'emily.r', '2024-11-05 10:00:00', 0),(29, 'New Movie Release', 'emily.r', '2024-11-05 10:30:00', 1),(30, 'Streaming Deals', 'emily.r', '2024-11-05 11:00:00', 1);

## 4.38 user_view_article Table

The user_view_article table logs whenever a user views an article on Netflix Tudum.

### 4.38.1 user_view_article DDL Script

```
CREATE TABLE user_view_article (
  viewarticle_id int(11) NOT NULL COMMENT 'View article ID',
  username varchar(20) NOT NULL COMMENT 'User name',
  article_title varchar(100) NOT NULL COMMENT 'Article title',
  article_view_date date NOT NULL COMMENT 'The date the article is viewed.'
) ENGINE=InnoDB DEFAULT CHARSET=utf8;
ALTER TABLE user_view_article
  ADD PRIMARY KEY (viewarticle_id),
  ADD KEY fk_viewarticle_username (username),
  ADD KEY fk_viewarticle_articletitle (article_title);
ALTER TABLE user_view_article
  MODIFY viewarticle_id int(11) NOT NULL AUTO_INCREMENT COMMENT 'View article ID', AUTO_INCREMENT=36;
ALTER TABLE user_view_article
  ADD CONSTRAINT fk_viewarticle_articletitle FOREIGN KEY (article_title) REFERENCES article_tudum (article_title),
  ADD CONSTRAINT fk_viewarticle_username FOREIGN KEY (username) REFERENCES user_account (username);
```

### 4.38.2 user_view_article table example data

```
INSERT INTO user_view_article (viewarticle_id, username, article_title, article_view_date) VALUES
(1, 'bob_the_builder', 'Sadie Sink on Max’s Trip to the Upside Down in ‘Stranger Things’ Season 4', '2024-02-11'),
```

(2, 'bob_the_builder', 'Sadie Sink on Max’s Trip to the Upside Down in ‘Stranger Things’ Season 4', '2024-11-15'),
(3, 'bob_the_builder', 'Sadie Sink on Max’s Trip to the Upside Down in ‘Stranger Things’ Season 4', '2024-12-05'),
(4, 'charlie123', 'The Duffer Brothers on Why ‘Stranger Things’ Season 4 Is Two Volumes', '2024-03-05'),
(5, 'charlie123', 'The Duffer Brothers on Why ‘Stranger Things’ Season 4 Is Two Volumes', '2024-11-17'),
(6, 'charlie123', 'The Duffer Brothers on Why ‘Stranger Things’ Season 4 Is Two Volumes', '2024-12-07'),
(7, 'alice_b', 'The Queen\'s Gambit Cast, News, Videos and more', '2024-01-10'),
(8, 'alice_b', 'The Queen\'s Gambit Cast, News, Videos and more', '2024-11-11'),
(9, 'alice_b', 'The Queen\'s Gambit Cast, News, Videos and more', '2024-12-03'),
(10, 'david_k', 'Who Is Eddie? Steve and Dustin’s ‘Stranger Things’ Friendship', '2024-04-01'),
(11, 'david_k', 'Who Is Eddie? Steve and Dustin’s ‘Stranger Things’ Friendship', '2024-11-18'),
(12, 'david_k', 'Who Is Eddie? Steve and Dustin’s ‘Stranger Things’ Friendship', '2024-12-10'),
(13, 'emily.r', '‘Stranger Things’ Animated Series Announced: What The Duffer Brothers Revealed So Far', '2024-05-15'),
(14, 'emily.r', '‘Stranger Things’ Animated Series Announced: What The Duffer Brothers Revealed So Far', '2024-11-20'),
(15, 'emily.r', '‘Stranger Things’ Animated Series Announced: What The Duffer Brothers Revealed So Far', '2024-12-12'),
(16, 'frank88', '‘Stranger Things’ Cast Red Carpet Premiere Photos', '2024-06-20'),
(17, 'frank88', '‘Stranger Things’ Cast Red Carpet Premiere Photos', '2024-11-22'),
(18, 'frank88', '‘Stranger Things’ Cast Red Carpet Premiere Photos', '2024-12-15'),
(19, 'gina_l', '‘Stranger Things’ Season 4 Easter Eggs', '2024-07-17'),
(20, 'gina_l', '‘Stranger Things’ Season 4 Easter Eggs', '2024-11-25'),
(21, 'gina_l', '‘Stranger Things’ Season 4 Easter Eggs', '2024-12-17'),
(22, 'harry_p', '‘Stranger Things’ Season 4 Episode Length Revealed', '2024-08-22'),
(23, 'harry_p', '‘Stranger Things’ Season 4 Episode Length Revealed', '2024-11-26'),
(24, 'harry_p', '‘Stranger Things’ Season 4 Episode Length Revealed', '2024-12-18'),

(25, 'jane_smith', ''Stranger Things' Season 4 First Look Photos', '2024-09-09'),
(26, 'bob_the_builder', 'Sadie Sink on Max's Trip to the Upside Down in 'Stranger Things' Season 4', '2024-10-05'),
(27, 'charlie123', 'The Duffer Brothers on Why 'Stranger Things' Season 4 Is Two Volumes', '2024-10-10'),
(28, 'alice_b', 'The Queen\'s Gambit Cast, News, Videos and more', '2024-10-15'),
(29, 'david_k', 'Who Is Eddie? Steve and Dustin's 'Stranger Things' Friendship', '2024-10-20'),
(30, 'frank88', ''Stranger Things' Cast Red Carpet Premiere Photos', '2024-10-25'),
(31, 'bob_the_builder', 'Sadie Sink on Max's Trip to the Upside Down in 'Stranger Things' Season 4', '2024-10-05'),
(32, 'charlie123', 'The Duffer Brothers on Why 'Stranger Things' Season 4 Is Two Volumes', '2024-10-10'),
(33, 'alice_b', 'The Queen\'s Gambit Cast, News, Videos and more', '2024-10-15'),
(34, 'david_k', 'Who Is Eddie? Steve and Dustin's 'Stranger Things' Friendship', '2024-10-20'),
(35, 'frank88', ''Stranger Things' Cast Red Carpet Premiere Photos', '2024-10-25');

### 4.39 user_view_content Table

The user_view_content table logs whenever a user decides to view a show on Netflix.

#### 4.39.1 user_view_content DDL Script

```
CREATE TABLE user_view_content (
  viewcontent_id int(11) NOT NULL COMMENT 'View Content ID',
  username varchar(20) NOT NULL COMMENT 'User name',
  content_id int(11) NOT NULL COMMENT 'Content ID',
  content_view_date date NOT NULL COMMENT 'The date the content is viewed.'
) ENGINE=InnoDB DEFAULT CHARSET=utf8;
ALTER TABLE user_view_content
  ADD PRIMARY KEY (viewcontent_id),
  ADD KEY fk_viewcontent_username (username),
  ADD KEY fk_viewcontent_contentid (content_id);
ALTER TABLE user_view_content
```

```
  MODIFY viewcontent_id int(11) NOT NULL AUTO_INCREMENT COMMENT
'View Content ID', AUTO_INCREMENT=31;
ALTER TABLE user_view_content
  ADD CONSTRAINT fk_viewcontent_contentid FOREIGN KEY (content_id)
REFERENCES content (content_id),
  ADD CONSTRAINT fk_viewcontent_username FOREIGN KEY (username)
REFERENCES user_account (username);
```

### 4.39.2 user_view_content  table example data

```
INSERT INTO user_view_content (viewcontent_id, username, content_id,
content_view_date) VALUES
(1, 'alice_b', 1, '2024-01-10'),(2, 'alice_b', 2, '2024-03-01'),(3, 'bob_the_builder', 2,
'2024-01-12'),(4, 'charlie123', 3, '2024-01-15'),(5, 'bob_the_builder', 4,
'2024-03-03'),(6, 'david_k', 4, '2024-01-18'),(7, 'emily.r', 5, '2024-01-20'),(8,
'charlie123', 6, '2024-03-05'),(9, 'frank88', 6, '2024-01-22'),(10, 'gina_l', 7,
'2024-01-25'),(11, 'david_k', 8, '2024-03-08'),(12, 'harry_p', 8, '2024-01-27'),(13,
'jane_smith', 9, '2024-01-30'),(14, 'emily.r', 10, '2024-03-10'),(15, 'john_doe', 10,
'2024-02-01'),(16, 'alice_b', 11, '2024-02-05'),(17, 'bob_the_builder', 12,
'2024-02-07'),(18, 'frank88', 12, '2024-03-12'),(19, 'charlie123', 13,
'2024-02-09'),(20, 'david_k', 14, '2024-02-11'),(21, 'gina_l', 14, '2024-03-15'),(22,
'emily.r', 15, '2024-02-13'),(23, 'frank88', 16, '2024-02-16'),(24, 'harry_p', 16,
'2024-03-17'),(25, 'gina_l', 17, '2024-02-18'),(26, 'harry_p', 18, '2024-02-20'),(27,
'jane_smith', 18, '2024-03-19'),(28, 'jane_smith', 19, '2024-02-22'),(29, 'john_doe',
20, '2024-02-24'),(30, 'john_doe', 20, '2024-03-21');
```

# 5 SQL Queries

## 5.1 Basic SQL Queries

### 5.1.1 Merchandise with Discounts

| **Report** | Merchandise with Discounts |
|---|---|
| **Objective** | Retrieve all merchandise items that currently have a discounted price, along with the percentage discount. |

**Query**

```
SELECT
   item_name,
   price,
   discounted_price,
   ROUND(((price - discounted_price) / price) * 100, 2) AS discount_percentage
FROM
   merchandise_item
WHERE
   discounted_price < price;
```

**Query Result**

| item_name<br>Item Name | price<br>Price | discounted_price<br>Discounted price. | discount_percentage |
|---|---|---|---|
| Squid Game Tracksuit | 49.99 | 44.99 | 10.00 |
| Stranger Things Mug | 15.99 | 12.99 | 18.76 |
| Squid Game Glass Bottle | 25.99 | 22.99 | 11.54 |

### 5.1.2 Profiles with Kids' Restrictions

| **Report** | Profiles with Kids' Restrictions |
|---|---|
| **Objective** | Find all user profiles designated as kids' profiles, including the associated maturity rating. |

**Query**

```
SELECT
   profile_name,
   username,
```

```
    maturity_rating
FROM
    profile
WHERE
    kids_profile_status = 1;
```

**Query Result**

| profile_name<br>Profile name | username<br>User name | maturity_rating<br>Maturity rating |
|---|---|---|
| LilBuddy | bob_the_builder | G |
| LilDreamer | emily.r | G |
| MiniExplorer | charlie123 | G |
| SpaceCadet | david_k | G |
| TinyAdventurer | frank88 | G |
| TinyTots | alice_b | G |
| WonderKid | gina_l | G |

### 5.1.3 Number of Active Subscriptions

| **Report** | Number of Active Subscriptions |
|---|---|
| **Objective** | Count the total number of active subscriptions in the database. |

**Query**

```
SELECT
    COUNT(subscription_id) AS active_subscriptions
FROM
    subscription
WHERE
    termination_date IS NULL;
```

**Query Result**

| active_subscriptions |
|---|
| 15 |

### 5.1.4 Maximum Downloads for a Single Game

| **Report** | Maximum Downloads for a Single Game |
|---|---|

| Objective | Find the highest number of downloads recorded for any game. |
|---|---|

**Query**

```
SELECT
   usa.user_name,
   COUNT(usa.showads_id) AS total_ads_shown
FROM
   user_shown_ads usa
GROUP BY
   usa.user_name
ORDER BY
   total_ads_shown DESC;
```

**Query Result**

| user_name | total_ads_shown ▾ 1 |
|---|---|
| charlie123 | 6 |
| emily.r | 6 |
| alice_b | 6 |
| jane_smith | 6 |
| bob_the_builder | 6 |

### 5.1.5 Coupon Usage Analysis

| Report | Coupon Usage Analysis |
|---|---|
| Objective | Determine the number of times each coupon code has been used |

**Query**

```
SELECT
   promo_code,
   COUNT(*) AS usage_count
FROM
   order_use_coupon
GROUP BY
   promo_code
ORDER BY
   usage_count DESC;
```

**Query Result**

| promo_code<br>The coupon code used in the order | usage_count ▼ 1 |
|---|---|
| SUMMER10 | 6 |
| WITCHER5 | 5 |
| STRANGE10 | 3 |
| BRIDGE15 | 3 |
| NEWYEAR5 | 3 |
| XMAS2024 | 3 |
| FALLDEAL | 2 |
| SQUID20 | 2 |
| BLACKFRI | 2 |
| SPRING25 | 1 |

### 5.1.6 Daily Ad Skipping Rate

| **Report** | Daily Ad Skipping Rate |
|---|---|
| **Objective** | Calculate the percentage of ads skipped on each date |

**Query**

```
SELECT
   DATE(show_datetime) AS ad_date,
   COUNT(CASE WHEN is_skipped = 1 THEN 1 END) * 100.0 / COUNT(*) AS skip_rate
FROM
   user_shown_ads
GROUP BY
   ad_date
ORDER BY
   ad_date ASC;
```

**Query Result**

| ad_date ▲ 1 | skip_rate |
|---|---|
| 2024-11-01 | 83.33333 |
| 2024-11-02 | 83.33333 |
| 2024-11-03 | 83.33333 |
| 2024-11-04 | 83.33333 |
| 2024-11-05 | 83.33333 |

### 5.1.7 User Activity on Articles

| **Report** | User Activity on Articles |
|---|---|
| **Objective** | Track the total number of articles viewed by each user. |
| **Query** | |

```
SELECT
   username,
   COUNT(*) AS total_articles_viewed
FROM
   user_view_article
GROUP BY
   username
ORDER BY
   total_articles_viewed DESC
LIMIT 10;
```

**Query Result**

| username<br>User name | total_articles_viewed ▼ 1 |
|---|---|
| david_k | 5 |
| alice_b | 5 |
| bob_the_builder | 5 |
| frank88 | 5 |
| charlie123 | 5 |
| harry_p | 3 |
| emily.r | 3 |
| gina_l | 3 |
| jane_smith | 1 |

### 5.1.8 Merchandise Stock Summary

| **Report** | Merchandise Stock Summary |
|---|---|
| **Objective** | Find the total and average stock quantities for all merchandise items. |
| **Query** | |

```
SELECT
   SUM(stock_quantity) AS total_stock,
   AVG(stock_quantity) AS average_stock
FROM
   merchandise_item;
```

**Query Result**

| total_stock | average_stock |
|---|---|
| 920 | 76.6667 |

### 5.1.9 Average Ad Duration

| **Report** | Average Ad Duration |
|---|---|
| **Objective** | Calculate the average duration of all advertisements |

**Query**

```
SELECT AVG(duration) AS average_duration
FROM ads;
```

**Query Result**

| average_duration |
|---|
| 30.0000 |

### 5.1.10 Top Performing Advertisements

| **Report** | Top Performing Advertisements |
|---|---|
| **Objective** | Identify the most frequently shown ads. |

**Query**

```
SELECT ads_title, COUNT(*) AS times_shown
FROM user_shown_ads
GROUP BY ads_title
ORDER BY times_shown DESC
LIMIT 5;
```

| Query Result |
|---|

| ads_title<br>The title of an ad. | times_shown ▾ 1 |
|---|---|
| New Movie Release | 5 |
| Gadget Promotion | 5 |
| Streaming Deals | 5 |
| Healthy Snacks | 5 |
| Holiday Travel | 5 |

### 5.1.11 Content View Count

| Report | Content View Count |
|---|---|
| **Objective** | Determine the most-watched content over time |

**Query**

```
SELECT content_id, COUNT(*) AS view_count
FROM user_view_content
GROUP BY content_id
ORDER BY view_count DESC
LIMIT 10;
```

**Query Result**

| content_id<br>Content ID | view_count ▾ 1 |
|---|---|
| 4 | 2 |
| 20 | 2 |
| 6 | 2 |
| 8 | 2 |
| 10 | 2 |
| 12 | 2 |
| 14 | 2 |
| 16 | 2 |
| 2 | 2 |
| 18 | 2 |

### 5.1.12 Plan popularity

| **Report** | Plan Popularity |
|---|---|
| **Objective** | Determine the most subscribed plans. |
| **Query** | |
| SELECT plan_name, COUNT(*) AS subscriber_count<br>FROM subscription<br>GROUP BY plan_name<br>ORDER BY subscriber_count DESC; | |
| **Query Result** | |

| plan_name<br>Plan name | subscriber_count ▼ 1 |
|---|---|
| PREMIUM | 8 |
| BASIC | 7 |
| STANDARD | 6 |
| BASICADS | 5 |
| MOBILE | 4 |

### 5.1.13 Count Kids Profiles

| **Report** | Count Kids Profiles |
|---|---|
| **Objective** | Find the total number of profiles marked as kids' profiles. |
| **Query** | |
| SELECT COUNT(*) AS total_kids_profiles<br>FROM profile<br>WHERE kids_profile_status = 1; | |
| **Query Result** | |

| total_kids_profiles |
|---|
| 7 |

### 5.1.14 Distribution of Apparel Colors

| **Report** | Distribution of Apparel Colors |
|---|---|
| **Objective** | Understand the variety and distribution of colors available for items. |
| **Query** | |
| SELECT color, COUNT(*) AS count<br>FROM apparel_color<br>GROUP BY color<br>ORDER BY count DESC; | |
| **Query Result** | |

| color<br>Available color of the item. | count ▾ 1 |
|---|---|
| Black | 2 |
| Red | 1 |
| Gray | 1 |
| Blue | 1 |
| White | 1 |
| Green | 1 |
| Yellow | 1 |

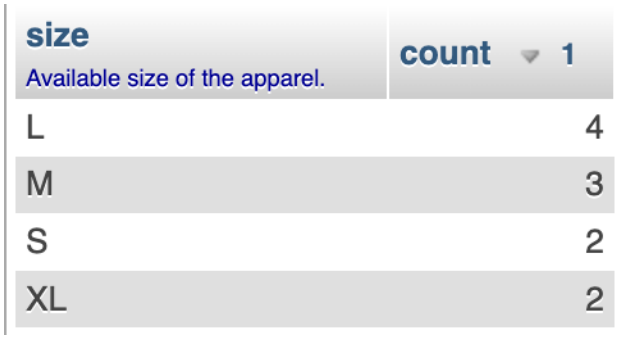
### 5.1.15 Profiles with PIN Enabled

| **Report** | Profiles with PIN Enabled |
|---|---|
| **Objective** | Count how many profiles have a PIN set. |
| **Query** | |
| SELECT COUNT(*) AS profiles_with_pin<br>FROM profile<br>WHERE pin IS NOT NULL; | |
| **Query Result** | |

| profiles_with_pin |
|---|
| 8 |

### 5.1.16 Sizes Availability Insights

| **Report** | Sizes Availability Insights |
|---|---|
| **Objective** | Determine the availability of different sizes to track inventory trends. |
| **Query** | |
| SELECT size, COUNT(*) AS count<br>FROM apparel_size<br>GROUP BY size<br>ORDER BY count DESC; | |
| **Query Result** | |



| size<br>Available size of the apparel. | count ▾ 1 |
|---|---|
| L | 4 |
| M | 3 |
| S | 2 |
| XL | 2 |

### 5.1.17 Top Advertisers by Total Ad Duration

| **Report** | Top Advertisers by Total Ad Duration |
|---|---|
| **Objective** | Determine the most subscribed plans. |
| **Query**<br>SELECT advertiser, SUM(duration) AS total_ad_duration<br>FROM ads GROUP<br>BY advertiser ORDER BY total_ad_duration DESC; | |
| **Query Result** | |

| ←T→ | advertiser<br>The advertiser of this ad. | total_ad_duration 1 |
|---|---|---|
| Edit Copy Delete | StreamPlus | 60 |
| Edit Copy Delete | GoGlobalTrips | 60 |
| Edit Copy Delete | InnovaTech | 45 |
| Edit Copy Delete | FashionHub | 45 |
| Edit Copy Delete | EcoMotors | 30 |
| Edit Copy Delete | FilmStudio | 30 |
| Edit Copy Delete | Foodies | 10 |
| Edit Copy Delete | ShopNow | 10 |
| Edit Copy Delete | FitStore | 5 |
| Edit Copy Delete | TechWorld | 5 |

### 5.1.18 Most Popular Content Maturity Ratings

| **Report** | Most Popular Content Maturity Ratings |
|---|---|
| **Objective** | Determine the count of content entries grouped by their maturity ratings to understand audience preferences |
| **Query** | |

```
SELECT maturity_rating, COUNT(*) AS content_count
FROM content
GROUP BY maturity_rating
ORDER BY content_count DESC;
```

**Query Result**

| maturity_rating<br>Maturity rating | content_count 1 |
|---|---|
| TV-MA | 9 |
| R | 6 |
| TV-14 | 3 |
| PG-13 | 2 |

### 5.1.19 Highest-Rated Merchandise

| **Report** | Highest-Rated Merchandise |
|---|---|
| **Objective** | Find the average rating for each merchandise item and rank them. |
| **Query** | |
| SELECT item_id, AVG(rating) AS average_rating<br>FROM merchandise_review<br>GROUP BY item_id<br>ORDER BY average_rating DESC; | |
| **Query Result** | |

| item_id<br>Item ID | average_rating 1 |
|---|---|
| 6 | 5.0000 |
| 7 | 5.0000 |
| 9 | 5.0000 |
| 11 | 5.0000 |
| 12 | 5.0000 |
| 2 | 5.0000 |
| 4 | 4.6667 |
| 5 | 4.3333 |
| 1 | 4.0000 |
| 8 | 3.6667 |
| 10 | 2.5000 |
| 3 | 2.5000 |

### 5.1.20 User Article Engagement

| **Report** | User Article Engagement |
|---|---|
| **Objective** | Count how many times each article has been viewed. |
| **Query** | |
| SELECT article_title, COUNT(*) AS view_count<br>FROM user_view_article<br>GROUP BY article_title<br>ORDER BY view_count DESC; | |
| **Query Result** | |

| article_title<br>Article title | view_count ▼ 1 |
|---|---|
| The Duffer Brothers on Why ‘Stranger Things’ Season 4 Is Two Volumes | 5 |
| The Queen's Gambit Cast, News, Videos and more | 5 |
| Who Is Eddie? Steve and Dustin’s ‘Stranger Things’ Friendship | 5 |
| Sadie Sink on Max’s Trip to the Upside Down in ‘Stranger Things’ Season 4 | 5 |
| ‘Stranger Things’ Cast Red Carpet Premiere Photos | 5 |
| ‘Stranger Things’ Season 4 Easter Eggs | 3 |
| ‘Stranger Things’ Season 4 Episode Length Revealed | 3 |
| ‘Stranger Things’ Animated Series Announced: What The Duffer Brothers Revealed So Far | 3 |
| ‘Stranger Things’ Season 4 First Look Photos | 1 |

### 5.1.21 Total Stock by Collection Category

| **Report** | Total Stock by Collection Category |
|---|---|
| **Objective** | Calculate the total stock available for each collection category to identify inventory levels. |

**Query**

```
SELECT
   collection_category,
   SUM(stock_quantity) AS total_stock
FROM
   merchandise_item
GROUP BY
   collection_category
ORDER BY
   total_stock DESC;
```

**Query Result**

| collection_category<br>The collection category this item belongs to. | total_stock ▼ 1 |
|---|---|
| The Witcher Collection | 290 |
| Stranger Things Collection | 250 |
| Squid Game Collection | 200 |
| Bridgerton Collection | 180 |

### 5.1.22 The Average Price of All Items in a Collection

| **Report** | The Average Price of All Items in a Collection |
|---|---|
| **Objective** | Find the average item price in each collection category. |

**Query**

```
SELECT
   collection_category,
   AVG(price) AS avg_price
FROM
   merchandise_item
GROUP BY
   collection_category
ORDER BY
   avg_price DESC;
```

**Query Result**

| collection_category<br>The collection category this item belongs to. | avg_price ▼ 1 |
|---|---|
| Squid Game Collection | 31.990000 |
| Bridgerton Collection | 23.656667 |
| The Witcher Collection | 20.990000 |
| Stranger Things Collection | 20.323333 |

### 5.1.23 Count the Number of Shows by Maturity Rating

| **Report** | Count the Number of Shows by Maturity Rating |
|---|---|
| **Objective** | This query provides an overview of the count of shows grouped by their maturity ratings. |

**Query**

```
SELECT
   maturity_rating,
   COUNT(*) AS show_count
FROM
   content
GROUP BY
   maturity_rating
ORDER BY
   show_count DESC;
```

**Query Result**

| maturity_rating<br>Maturity rating | show_count ▼ 1 |
|---|---|
| TV-MA | 9 |
| R | 6 |
| TV-14 | 3 |
| PG-13 | 2 |

### 5.1.24 Number of articles by category

| **Report** | Number of articles by category |
|---|---|
| **Objective** | Count the number of articles by category. |
| **Query** | |

```
SELECT category, COUNT(*) AS article_count
FROM article_tudum
GROUP BY category
ORDER BY article_count DESC;
```

**Query Result**

| category<br>The category of an article. | article_count ▼ 1 |
|---|---|
| news | 4 |
| meet the cast | 3 |
| deep dive | 2 |
| explainer | 1 |

## 5.2 Advanced SQL Queries

### 5.2.1 Top-Rated Merchandise Items

| **Report** | Top-Rated Merchandise Items |
|---|---|
| **Objective** | Find the top 5 merchandise items with the highest average customer rating. |
| **Query** | |
| SELECT mi.item_name, ROUND(AVG(mr.rating), 2) AS average_rating,<br>COUNT(mr.review_id) AS total_reviews<br>FROM merchandise_item mi<br>JOIN merchandise_review mr ON mi.item_id = mr.item_id GROUP BY mi.item_name ORDER BY average_rating DESC, total_reviews DESC LIMIT 5; | |
| **Query Result** | |

| item_name | average_rating 1 | total_reviews 2 |
|---|---|---|
| The Witcher Tumbler | 5.00 | 3 |
| Stranger Things Funko Pop | 5.00 | 3 |
| Bridgerton Crown Replica | 5.00 | 2 |
| Squid Game Figure | 5.00 | 2 |
| The Witcher Hoodie | 5.00 | 2 |

### 5.2.2 Game Download Count

| **Report** | Game Download Count |
|---|---|
| **Objective** | Identify how many times each game has been downloaded. |
| **Query** | |
| SELECT g.game_name, COUNT(udg.download_id) AS download_count<br>FROM game g<br>JOIN user_download_game udg ON g.game_name = udg.game_name<br>GROUP BY g.game_name<br>ORDER BY download_count DESC; | |
| **Query Result** | |

| game_name | download_count ▼ 1 |
|---|---|
| Heads Up! | 4 |
| Stranger Things: 1984 | 3 |
| Moonlighter | 3 |
| Hextech Mayhem | 3 |
| Asphalt Xtreme | 3 |
| Shatter Remastered | 3 |
| Kentucky Route Zero | 3 |
| Oxenfree | 3 |
| Into the Breach | 3 |
| Exploding Kittens | 3 |

### 5.2.3 Top-Selling Merchandise by Revenue

| **Report** | Top-Selling Merchandise by Revenue |
|---|---|
| **Objective** | Identify the top 5 merchandise items that generated the most revenue, including their total sales revenue. |
| **Query** | |

```
SELECT
   mi.item_name,
   SUM(o.quantity * mi.price) AS total_revenue
FROM
   merchandise_item mi
JOIN
   order_contains_item o ON mi.item_id = o.item_id
GROUP BY
   mi.item_name
ORDER BY
   total_revenue DESC
LIMIT 5;
```

**Query Result**

| item_name | total_revenue ▾ 1 |
|---|---|
| Stranger Things T-Shirt | 359.88 |
| The Witcher Hoodie | 319.92 |
| Squid Game Tracksuit | 299.94 |
| Bridgerton Crown Replica | 249.90 |
| Squid Game Glass Bottle | 207.92 |

### 5.2.4 Recently Released TV Shows

| **Report** | Most Popular Games by Downloads |
|---|---|
| **Objective** | Retrieve the titles of TV shows released in the last 2 years and their series status. |

**Query**

```
SELECT
   c.content_title AS tv_show_title,
   ts.series_status,
   c.release_date
FROM
   content c
JOIN
   tv_show ts ON c.content_id = ts.content_id
WHERE
   c.release_date >= DATE_ADD(CURRENT_DATE(), INTERVAL -2 YEAR);
```

**Query Result**

| tv_show_title | series_status | release_date |
|---|---|---|
| A Killer Paradox | Ongoing | 2024-01-15 |
| One Piece | Ongoing | 2023-08-31 |

### 5.2.5 Retrieve Apparel Colors and Materials

| **Report** | Retrieve Apparel Colors and Materials |
|---|---|
| **Objective** | To display all apparel items along with their respective colors and materials, providing a complete overview of available products. |

**Query**

```
SELECT a.item_id, a.material, ac.color
FROM apparel a
JOIN apparel_color ac ON a.item_id = ac.item_id;
```

**Query Result**

| item_id | material | color |
|---|---|---|
| 1 | 100% Cotton | Black |
| 1 | 100% Cotton | Red |
| 2 | Polyester/Cotton Blend | Black |
| 2 | Polyester/Cotton Blend | Gray |
| 3 | Organic Cotton | Blue |
| 3 | Organic Cotton | White |
| 4 | Polyester | Green |
| 4 | Polyester | Yellow |

### 5.2.6 Find all apparel items available in "Black" color.

| | |
|---|---|
| **Report** | Find all apparel items available in "Black" color. |
| **Objective** | To identify all apparel items that are available in the "Black" color, useful for targeted inventory queries or promotional campaigns. |

**Query**

```
SELECT a.item_id, a.material
FROM apparel a
JOIN apparel_color ac ON a.item_id = ac.item_id
WHERE ac.color = 'Black';
```

**Query Result**

| item_id | material |
|---|---|
| 1 | 100% Cotton |
| 2 | Polyester/Cotton Blend |

### 5.2.7 List apparel items and sizes where the material is 'Organic Cotton'.

| | |
|---|---|
| **Report** | List apparel items and sizes where the material is 'Organic Cotton'. |
| **Objective** | To list all apparel items made from "Organic Cotton" along with their available sizes, useful for eco-conscious customer segments. |

**Query**

```
SELECT a.item_id, a.material, s.size
FROM apparel a
JOIN apparel_size s ON a.item_id = s.item_id
WHERE a.material = 'Organic Cotton';
```

**Query Result**

| item_id | material | size |
|---|---|---|
| 3 | Organic Cotton | L |
| 3 | Organic Cotton | M |
| 3 | Organic Cotton | S |

### 5.2.8 List All Apparel Items with Color and Size

| | |
|---|---|
| **Report** | List All Apparel Items with Color and Size |
| **Objective** | Identify ads that relate to articles (via content IDs or similar relationships). |

**Query**

```
SELECT ap.item_id, ap.material, ac.color, asz.size
FROM apparel ap
JOIN apparel_color ac ON ap.item_id = ac.item_id
JOIN apparel_size asz ON ap.item_id = asz.item_id;
```

**Query Result**

| item_id | material | color | size |
|---|---|---|---|
| 1 | 100% Cotton | Black | L |
| 1 | 100% Cotton | Black | M |
| 1 | 100% Cotton | Black | S |
| 1 | 100% Cotton | Red | L |
| 1 | 100% Cotton | Red | M |
| 1 | 100% Cotton | Red | S |
| 2 | Polyester/Cotton Blend | Black | L |
| 2 | Polyester/Cotton Blend | Black | M |
| 2 | Polyester/Cotton Blend | Black | XL |
| 2 | Polyester/Cotton Blend | Gray | L |
| 2 | Polyester/Cotton Blend | Gray | M |
| 2 | Polyester/Cotton Blend | Gray | XL |
| 3 | Organic Cotton | Blue | L |
| 3 | Organic Cotton | Blue | M |
| 3 | Organic Cotton | Blue | S |
| 3 | Organic Cotton | White | L |
| 3 | Organic Cotton | White | M |
| 3 | Organic Cotton | White | S |
| 4 | Polyester | Green | L |
| 4 | Polyester | Green | XL |
| 4 | Polyester | Yellow | L |
| 4 | Polyester | Yellow | XL |

### 5.2.9 Most popular TV show by views.

| **Report** | Most popular TV show by views. |
|---|---|
| **Objective** | Identify the TV show with the most views in descending order. |
| **Query** | |
| SELECT c.content_title, COUNT(v.viewcontent_id) AS view_count<br>FROM tv_show t<br>JOIN user_view_content v ON t.content_id = v.content_id<br>JOIN content c ON t.content_id = c.content_id<br>GROUP BY t.content_id<br>ORDER BY view_count DESC; | |
| **Query Result** | |

| content_title | view_count ▼ 1 |
|---|---|
| The Crown | 2 |
| One Piece | 2 |
| The Lincoln Lawyer | 2 |
| Squid Game | 2 |
| Money Heist | 2 |
| Bridgerton | 1 |
| Stranger Things | 1 |
| The Queen's Gambit | 1 |
| A Killer Paradox | 1 |
| The Witcher | 1 |

### 5.2.10 Users Who Viewed Content Across Multiple Dates

| **Report** | Users Who Viewed Content Across Multiple Dates |
|---|---|
| **Objective** | List users who viewed content on more than one distinct date. |

**Query**

```
SELECT v.username, COUNT(DISTINCT v.content_view_date) AS view_dates
FROM user_view_content v
GROUP BY v.username
HAVING view_dates > 1;
```

**Query Result**

| username | view_dates |
|---|---|
| alice_b | 3 |
| bob_the_builder | 3 |
| charlie123 | 3 |
| david_k | 3 |
| emily.r | 3 |
| frank88 | 3 |
| gina_l | 3 |
| harry_p | 3 |
| jane_smith | 3 |
| john_doe | 3 |

### 5.2.11 Users Who Interacted with Content or Games

| **Report** | Users Who Interacted with Content or Games |
|---|---|
| **Objective** | Identify all users who either viewed content or downloaded games. |

**Query**

```
SELECT username FROM user_view_content
UNION
SELECT username FROM user_download_game;
```

**Query Result**

| username |
|---|
| alice_b |
| bob_the_builder |
| charlie123 |
| david_k |
| emily.r |
| frank88 |
| gina_l |
| harry_p |
| jane_smith |
| john_doe |

### 5.2.12 Gender Percentage of Viewers On Each Content

| | |
|---|---|
| **Report** | Gender Percentage of Viewers On Each Content |
| **Objective** | Display the gender percentage of viewers on each content. Note that only users subscribed to ad-enabled plans will have their gender data available. |

**Query**

```
SELECT
    c.content_title,
    ROUND(SUM(CASE WHEN ua.gender = 'M' THEN 1 ELSE 0 END) * 100.0 /
COUNT(ua.username), 2) AS male_percentage,
    ROUND(SUM(CASE WHEN ua.gender = 'F' THEN 1 ELSE 0 END) * 100.0 /
COUNT(ua.username), 2) AS female_percentage
FROM
    user_view_content uvc
JOIN
    user_account ua ON uvc.username = ua.username
JOIN
    content c ON uvc.content_id = c.content_id
WHERE
    ua.gender IS NOT NULL
GROUP BY
    c.content_title
ORDER BY
    c.content_title;
```

**Query Result**

| content_title | male_percentage | female_percentage |
|---|---|---|
| Bridgerton | 0.00 | 100.00 |
| Don't Look Up | 0.00 | 100.00 |
| Enola Holmes | 100.00 | 0.00 |
| Marriage Story | 0.00 | 100.00 |
| One Piece | 100.00 | 0.00 |
| Squid Game | 50.00 | 50.00 |
| Stranger Things | 0.00 | 100.00 |
| The Harder They Fall | 100.00 | 0.00 |
| The Irishman | 0.00 | 100.00 |
| The Lincoln Lawyer | 100.00 | 0.00 |
| The Old Guard | 0.00 | 100.00 |
| The Power of the Dog | 0.00 | 100.00 |
| The Witcher | 0.00 | 100.00 |

### 5.2.13 Find Advertisers Whose Ads Were Shown to Users

| **Report** | Find Advertisers Whose Ads Were Shown to Users |
|---|---|
| **Objective** | Determine which advertisers reached users through their ad campaigns. |
| **Query** | |
| SELECT DISTINCT a.advertiser, u.user_name<br>FROM ads a<br>JOIN user_shown_ads u ON a.ads_title = u.ads_title; | |
| **Query Result** | |

| advertiser | user_name |
|---|---|
| EcoMotors | alice_b |
| EcoMotors | charlie123 |
| EcoMotors | jane_smith |
| EcoMotors | bob_the_builder |
| EcoMotors | emily.r |
| TechWorld | alice_b |
| TechWorld | charlie123 |
| TechWorld | jane_smith |
| TechWorld | bob_the_builder |
| TechWorld | emily.r |
| Foodies | alice_b |
| Foodies | charlie123 |
| Foodies | jane_smith |
| Foodies | bob_the_builder |
| Foodies | emily.r |
| GoGlobalTrips | alice_b |
| GoGlobalTrips | charlie123 |
| GoGlobalTrips | jane_smith |
| GoGlobalTrips | bob_the_builder |
| GoGlobalTrips | emily.r |
| FilmStudio | alice_b |
| FilmStudio | charlie123 |
| FilmStudio | jane_smith |
| FilmStudio | bob_the_builder |
| FilmStudio | emily.r |

### 5.2.14 Find Drinkware Items Ordered and Their Sizes

| **Report** | Find Drinkware Items Ordered and Their Sizes |
|---|---|
| **Objective** | Extract detailed information on ordered drinkware items for inventory or sales analysis. |
| **Query** | |
| SELECT mi.item_name, d.capacity, d.upper_diameter, d.bottom_diameter<br>FROM merchandise_item mi<br>JOIN drinkware d ON mi.item_id = d.item_id<br>JOIN order_contains_item oci ON mi.item_id = oci.item_id; | |
| **Query Result** | |

| item_name | capacity | upper_diameter | bottom_diameter |
|---|---|---|---|
| Stranger Things Mug | 350 | 3 | 3 |
| Stranger Things Mug | 350 | 3 | 3 |
| Stranger Things Mug | 350 | 3 | 3 |
| Stranger Things Mug | 350 | 3 | 3 |
| Bridgerton Wine Glass | 450 | 3 | 2 |
| Bridgerton Wine Glass | 450 | 3 | 2 |
| Bridgerton Wine Glass | 450 | 3 | 2 |
| Bridgerton Wine Glass | 450 | 3 | 2 |
| The Witcher Tumbler | 500 | 3 | 3 |
| The Witcher Tumbler | 500 | 3 | 3 |
| The Witcher Tumbler | 500 | 3 | 3 |
| The Witcher Tumbler | 500 | 3 | 3 |
| Squid Game Glass Bottle | 750 | 3 | 2 |
| Squid Game Glass Bottle | 750 | 3 | 2 |
| Squid Game Glass Bottle | 750 | 3 | 2 |
| Squid Game Glass Bottle | 750 | 3 | 2 |

### 5.2.15 List All Directors for Content in the 'Action' Genre

| | |
|---|---|
| **Report** | List All Directors for Content in the 'Action' Genre |
| **Objective** | Identify directors associated with a particular genre for genre-specific promotions or trends analysis. |

**Query**

```
SELECT cd.director, g.genre_name
FROM content_director cd
JOIN genre_contain_content gc ON cd.content_id = gc.content_id
JOIN genre g ON gc.genre_id = g.genre_id
WHERE g.genre_name = 'Action';
```

**Query Result**

| director | genre_name |
|---|---|
| Rawson Marshall Thurber | Action |
| Gina Prince-Bythewood | Action |
| Martin Scorsese | Action |

### 5.2.16 Display Users Who Subscribed to Specific Plans

| Report | Display Users Who Subscribed to Specific Plans |
|---|---|
| **Objective** | Link user accounts to their subscription details to understand plan distribution and revenue streams. |

**Query**

```
SELECT u.username, s.plan_name, p.monthly_price
FROM user_account u
JOIN subscription s ON u.username = s.username
JOIN plan p ON s.plan_name = p.plan_name;
```

**Query Result**

| username | plan_name | monthly_price |
|---|---|---|
| alice_b | BASIC | 9.99 |
| charlie123 | BASIC | 9.99 |
| emily.r | BASIC | 9.99 |
| frank88 | BASIC | 9.99 |
| harry_p | BASIC | 9.99 |
| alice_b | BASIC | 9.99 |
| david_k | BASIC | 9.99 |
| alice_b | BASICADS | 6.99 |
| charlie123 | BASICADS | 6.99 |
| jane_smith | BASICADS | 6.99 |
| bob_the_builder | BASICADS | 6.99 |
| emily.r | BASICADS | 6.99 |
| bob_the_builder | MOBILE | 3.99 |
| david_k | MOBILE | 3.99 |
| frank88 | MOBILE | 3.99 |
| bob_the_builder | MOBILE | 3.99 |
| bob_the_builder | PREMIUM | 19.99 |
| david_k | PREMIUM | 19.99 |
| gina_l | PREMIUM | 19.99 |
| jane_smith | PREMIUM | 19.99 |
| john_doe | PREMIUM | 19.99 |
| alice_b | PREMIUM | 19.99 |
| charlie123 | PREMIUM | 19.99 |
| david_k | PREMIUM | 19.99 |
| emily.r | STANDARD | 15.99 |

### 5.2.17 Advertisers with Most Ads Shown

| Report | Advertisers with Most Ads Shown |
|---|---|
| **Objective** | Identify advertisers whose ads were shown most frequently to users. |

**Query**

```
SELECT a.advertiser, COUNT(usa.ads_title) AS ads_shown_count
FROM ads a
JOIN user_shown_ads usa ON a.ads_title = usa.ads_title
GROUP BY a.advertiser
ORDER BY ads_shown_count DESC;
```

**Query Result**

| advertiser | ads_shown_count ▼ 1 |
|---|---|
| FilmStudio | 5 |
| TechWorld | 5 |
| StreamPlus | 5 |
| Foodies | 5 |
| GoGlobalTrips | 5 |
| EcoMotors | 5 |

### 5.2.18 Top 5 Rated Merchandise Items

| **Report** | Top 5 Rated Merchandise Items |
|---|---|
| **Objective** | Find the top 5 highest-rated merchandise items based on user reviews. |

**Query**

```
SELECT
   mi.collection_category,
   SUM(mi.price) AS total_revenue
FROM
   merchandise_item mi
JOIN
   merchandise_order mo ON mi.item_id = mo.order_id
WHERE
   mo.status = 'completed'
GROUP BY
   mi.collection_category
ORDER BY
   total_revenue DESC;
```

**Query Result**

| collection_category | total_revenue ▼ 1 |
|---|---|
| Squid Game Collection | 49.99 |
| Stranger Things Collection | 44.98 |
| Bridgerton Collection | 44.98 |
| The Witcher Collection | 9.99 |

### 5.2.19 Revenue from Discounted Merchandise

| **Report** | Revenue from Discounted Merchandise |
|---|---|
| **Objective** | Calculate the total revenue generated from merchandise sold at discounted prices. |

**Query**

```
SELECT SUM(oi.quantity * mi.discounted_price) AS discounted_revenue
FROM order_contains_item oi
JOIN merchandise_item mi ON oi.item_id = mi.item_id
WHERE mi.discounted_price IS NOT NULL;
```

**Query Result**

| discounted_revenue |
|---|
| 557.78 |

### 5.2.20 Total View Count of Each Genre

| **Report** | Total View Count of Each Genre |
|---|---|
| **Objective** | List the genres and the total view count under each one. |

**Query**

```
SELECT gc.genre_name, COUNT(uvc.content_id) AS view_count
FROM user_view_content uvc
JOIN genre_contain_content gcc ON uvc.content_id = gcc.content_id
JOIN genre gc ON gcc.genre_id = gc.genre_id
GROUP BY gc.genre_name
ORDER BY view_count DESC;
```

**Query Result**

| genre_name | view_count ▼ 1 |
|---|---|
| Drama | 24 |
| Thriller | 8 |
| Crime | 7 |
| Adventure | 7 |
| Comedy | 4 |
| Action | 4 |
| Mystery | 3 |
| Romance | 3 |
| Fantasy | 3 |
| Historical | 2 |
| Western | 2 |
| Horror | 1 |

### 5.2.21 Active Subscriptions by Plan

| **Report** | Active Subscriptions by Plan |
|---|---|
| **Objective** | Count the number of active subscriptions for each plan and the total revenue from those subscriptions. |

**Query**

```
SELECT
    p.plan_name,
    COUNT(s.subscription_id) AS active_subscriptions,
    SUM(p.monthly_price) AS total_monthly_revenue
FROM
    plan p
JOIN
    subscription s ON p.plan_name = s.plan_name
WHERE
    s.termination_date IS NULL OR s.termination_date > CURRENT_DATE()
GROUP BY
    p.plan_name
ORDER BY
```

| active_subscriptions DESC; |
|---|
| **Query Result** |

| plan_name | active_subscriptions ▼ 1 | total_monthly_revenue |
|---|---|---|
| PREMIUM | 7 | 139.93 |
| BASICADS | 5 | 34.95 |
| STANDARD | 2 | 31.98 |
| MOBILE | 1 | 3.99 |

### 5.2.22 Find the most-watched maturity rating category.

| **Report** | Find the most-watched maturity rating category. |
|---|---|
| **Objective** | To determine the most-watched maturity rating category based on user viewing data |
| **Query** | |

```
SELECT c.maturity_rating,
COUNT(uvc.viewcontent_id)
AS total_views FROM content
AS c JOIN user_view_content
AS uvc ON c.content_id = uvc.content_id
GROUP BY c.maturity_rating
ORDER BY total_views DESC LIMIT 1;
```

**Query Result**

| maturity_rating | total_views |
|---|---|
| TV-MA | 13 |

### 5.2.23 Find the total number of views for each piece of content.

| **Report** | Find the total number of views for each piece of content. |
|---|---|
| **Objective** | To identify the total number of views for each piece of content, providing insights into content performance and popularity. |
| **Query** | |

```
SELECT c.content_title,
COUNT(uvc.viewcontent_id) AS total_views
FROM content AS c
LEFT JOIN user_view_content AS uvc ON c.content_id = uvc.content_id
GROUP BY c.content_title
ORDER BY total_views DESC;
```

| Query Result |
|---|

| content_title | total_views ▼ 1 |
|---|---|
| One Piece | 2 |
| The Harder They Fall | 2 |
| Enola Holmes | 2 |
| The Lincoln Lawyer | 2 |
| Squid Game | 2 |
| The Old Guard | 2 |
| Money Heist | 2 |
| Bird Box | 2 |
| The Crown | 2 |
| Don't Look Up | 2 |
| Stranger Things | 1 |
| Red Notice | 1 |
| The Queen's Gambit | 1 |
| The Killer | 1 |
| The Irishman | 1 |
| A Killer Paradox | 1 |
| The Witcher | 1 |
| Marriage Story | 1 |
| Bridgerton | 1 |
| The Power of the Dog | 1 |

**5.2.24 The number of cast members on each content title.**

| **Report** | The number of cast members on each content title. |
|---|---|
| **Objective** | To determine the number of cast members associated with each content title. |
| **Query** | |

```
SELECT c.content_title,
COUNT(uvc.viewcontent_id) AS total_views
FROM content AS c
LEFT JOIN user_view_content AS uvc ON c.content_id = uvc.content_id
GROUP BY c.content_title
ORDER BY total_views DESC;
```

| **Query Result** |
|---|

| content_title | total_cast |
|---|---|
| Marriage Story | 3 |
| The Crown | 3 |
| Bird Box | 3 |
| Stranger Things | 3 |
| The Power of the Dog | 3 |
| One Piece | 3 |
| Don't Look Up | 3 |
| The Queen's Gambit | 3 |
| Red Notice | 3 |
| Squid Game | 3 |
| Enola Holmes | 3 |
| The Witcher | 3 |
| The Irishman | 3 |
| Money Heist | 3 |
| The Old Guard | 3 |
| Bridgerton | 3 |
| The Lincoln Lawyer | 2 |
| The Harder They Fall | 2 |
| A Killer Paradox | 2 |
| The Killer | 2 |